

Do Environmental Provisions in Trade Agreements Reshape Export Composition?

Evidence from the Universe of Chinese Customs Transactions, 2000–2015

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August 2026 — **preliminary draft**

Abstract

Environmental Provisions (EPs) have been increasingly embedded in Preferential Trade Agreements (PTAs) over the last two decades, with the expectation that they will play an even more important role in future agreements. However, empirical evidence on their trade effects relies almost entirely on aggregate bilateral flows. This paper asks whether EPs can shift export composition towards green goods and away from dirty goods, using a firm-product-destination-year panel for the universe of Chinese customs transactions spanning from 2000 to 2015. A triple-difference design compares green and dirty products against neutral counterparts within each firm-destination-year combination, with firm-destination-year fixed effects accounting for everything constant within each firm-market-year triple. The results indicate that there is no effect on the green margin, while the dirty margin is less clear-cut. The inference on the green margin is robust to wild cluster bootstrap and permutation inference; that for the dirty margin is not. What emerges is that the content of environmental provisions matters more than their mere presence. The PTAs signed by China over 2000–2015 contained mostly cooperation language, which makes our results consistent with the existing literature, where significant effects are driven by specific and enforceable clauses rather than loose language.

1 Introduction

Over the last decades, the number of preferential trade agreements (PTAs) has increased substantially, and many now include environmental provisions (EPs) or other forms of environmental language (Mattoo, Rocha, and Ruta, 2020; Morin, Pauwelyn, and Hollway, 2017). Because climate change is likely to become an increasingly important political concern, and given the evidence that global trade may play a role in environmental decay (Zhang et al., 2017), environmental content in trade agreements is expected to feature more frequently and more prominently in trade policy, as it may serve as a possible tool to address the challenges posed by climate change (Frankel, 2009). Accordingly, the relationship between trade and the environment has attracted growing attention in the literature (Copeland, Shapiro, and Taylor, 2022), yet whether environmental provisions in trade agreements have a genuine effect on trade flows is still an open empirical question (Felbermayr, Peterson, and Wanner, 2025; Gutsch et al., 2025). This paper contributes to this literature by asking whether the environmental provisions included in preferential trade agreements signed by China between 2000 and 2015 shifted the composition of its exports toward greener products and away from dirtier ones, or whether those provisions left trade flows largely unaffected.

Most of the existing literature on the trade effects of environmental provisions in trade agreements comes from studies relying on aggregate bilateral trade flows, which makes it difficult to understand whether EPs cause a compositional shift in traded goods. In what the authors claim to be the first article to investigate the trade effects of environmental provisions in PTAs, Brandi et al. (2020) find that EPs reduce dirty exports and increase green exports when regulations are specific and enforceable, especially for developing countries. On the other hand, Berger et al. (2020) use a gravity panel to show that membership in PTAs with more environmental provisions is associated with lower bilateral exports overall, with the effect being concentrated in South-North trade flows and not robust to PPML. Specific and enforceable clauses are also those that drive the effect in Abman, Lundberg, and Ruta, 2024, who find that forest preservation and biodiversity clauses in PTAs reduce deforestation in partner countries relative to those that do not include such clauses. Taken together, the evidence suggests that it is the content of the specific provisions and chapters that matters, rather than their mere presence in trade agreements. This paper brings the question to the largest exporter in the world, asking whether the environmental provisions in China's PTAs between 2000 and 2015 had any effect on the composition of its exports, and whether the content of those provisions matters for trade flows.

Trying to answer this empirical question comes with some challenges. The first and most obvious is that countries do not sign trade agreements at random, and the signing of trade

agreements is likely to be correlated with pre-existing relationships of some sort. Therefore, any comparison of export flows to PTA partners versus non-partners risks confounding the effect of environmental provisions with pre-existing differences in trade relationships. A second issue is that aggregate bilateral trade flows cannot distinguish a reallocation of export composition toward greener goods from an expansion or contraction of trade that happens to coincide with agreement entry into force. To address the empirical question, this study uses the universe of Chinese customs transactions between 2000 and 2015, a 45-million-observation firm-product-destination-year panel with granularity at the HS6 product level, which makes it possible to track the same firm exporting the same product to multiple destinations simultaneously, and to observe how those flows diverge when one destination is covered by a preferential agreement and another is not. This design isolates the effect of environmental provisions on trade composition from other confounding factors.

Over the study period, China signed 14 PTAs with 25 destination economies, with environmental content ranging from a handful of generic cooperation clauses to dedicated chapters. Data on the content of the environmental provisions come from two independent sources, the World Bank Deep Trade Agreements database (Hofmann, Osnago, and Ruta, 2017) and the TREND database (Morin, Pauwelyn, and Hollway, 2017). The former provides a count of environmental-law provisions in force with each destination at each point in time, while the latter provides a more detailed coding of the content of those provisions, including sub-indices covering areas such as green market access and enforcement stringency. Using two independent codings guards against idiosyncrasies of either scheme and allows to check for robustness of the results. We identify green goods on the basis of the OECD Combined List of Environmental Goods (CLEG) (Sauvage, 2014), while dirty goods are identified through a standard pollution-intensity classification that maps polluting industries to HS6 product codes, as developed in Mani and Wheeler, 1998. Everything else serves as the neutral comparison group.

These three product categories, green, dirty and neutral, represent the foundation of the empirical strategy. The empirical design applied in this study is a comparison of how the exports of green and dirty products react to environmental provisions, relative to neutral products, within the same firm, destination and year. A triple-difference with a rich set of fixed effects, including firm-destination-year fixed effects, absorbs everything that is constant within a firm’s relationship with a specific destination market in a given year, including the agreement itself and any aggregate demand or supply shocks at the bilateral level. Because EP enter into force with the agreement itself and rarely change their depth or content afterwards, EP depth and PTA depth are highly correlated, as agreements with more EP content tend to be also deeper agreements overall. Therefore, all specifications control for PTA depth

separately, so that the coefficient on environmental provisions reflects proper environmental content. In this design, identification comes from the differential sensitivity of green and dirty products to environmental provisions, relative to neutral products, which serve as a within-cell control since they are not expected to be affected by environmental provisions as much as green and dirty products are.

The results indicate that there is no effect on the green margin, so the export of green goods relative to neutral ones is not affected by the depth of environmental provisions in China’s PTAs. This result is stable across six different estimation designs that vary the unit of observation, the measurement instrument, and the comparison group. The baseline uses the full firm-product-destination-year panel; a collapsed specification aggregates the same data to HS6-destination-year cells; and both are replicated using the TREND provision count as an alternative measure of EP depth to that offered by the World Bank Deep Trade Agreements database. Further robustness checks restrict the sample to CEM-matched destinations and to the common-support overlap region. Across all specifications, we find no statistically significant effect of environmental provisions on the export of green goods relative to neutral ones. Furthermore, these results are robust to wild cluster bootstrap inference (Cameron, Gelbach, and Miller, 2008) and permutation inference (Young, 2019). On the other hand, results for the dirty margin are less clear-cut. While we find no effect in the full panel, the collapsed panel shows a negative and significant coefficient, suggesting that while EPs may not change firm-specific behaviour, they may reduce the overall market-level export of dirty goods relative to neutral ones. However, this result does not survive more robust inference. Wild cluster bootstrap raises the p -value from below 0.001 to 0.07; a permutation test, which asks whether the same pattern would emerge by chance if EP depth were randomly reassigned across destinations, raises it further to 0.28.

The findings show that environmental provisions in China’s PTAs between 2000 and 2015 had no measurable effect on export composition at the firm-product-destination-year level. This result, however, should not be taken as conclusive evidence that EPs in trade agreements have no effect on trade flows at all, but rather reflects the nature of the specific provisions themselves. During the period under analysis, the environmental content of China’s agreements consisted almost entirely of soft cooperation language, rather than specific and enforceable clauses backed by dispute settlement mechanisms. The types of clauses that have been shown to produce significant trade outcomes in other contexts, such as tariff concessions on environmental goods and binding emissions or process standards, are largely absent from these agreements. In this sense, the null result is not a puzzle, but rather the expected outcome given the existing literature’s own heterogeneity results, applied to a setting where the provisions lack the specificity and enforceability that drive the effects doc-

umented elsewhere. The fact that the null result is robust to a variety of estimation designs and inference methods further supports the conclusion that the lack of effect is not due to methodological shortcomings, but rather reflects the substantive nature of the environmental provisions in China’s PTAs during this period. Moreover, the findings are informative for policy makers, as they suggest that the mere inclusion of environmental provisions in trade agreements may not be enough to achieve the desired trade outcomes if those provisions are just loose cooperation language, even in the case of the largest exporter in the world. Instead, the design and content of the provisions themselves, including specificity and enforceability, are likely to be crucial for achieving meaningful trade effects.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data and the agreements. Section 4 presents the empirical strategy. Section 5 reports the results, with robustness analysis integrated as subsections. Section 6 concludes.

2 Related Literature

A large body of evidence documents that preferential trade agreements increase trade between member countries. Using a panel gravity model that accounts for the endogeneity of the decision to sign an agreement, Baier and Bergstrand (2007) estimate that an FTA roughly doubles bilateral trade over a ten-year period, a finding substantially larger than earlier cross-sectional estimates that did not control for selection into agreements. Subsequent work has refined this result by showing that the overall trade effect depends on agreement characteristics: the magnitude varies with the depth and enforceability of the agreement, rather than being a uniform response to preferential market access (Baier and Bergstrand, 2009), and trade effects are heterogeneous along several other dimensions, including the type of members involved and the timing of liberalization (Medvedev, 2010). Surveying this literature more broadly, Freund and Ornelas (2010) argue that the distinction between shallow agreements, which primarily reduce tariffs, and deep agreements, which extend to regulatory harmonization, investment, services, and other behind-the-border provisions, is central to understanding their economic effects.

This distinction has been formalized through provision-level coding efforts. The World Bank’s Deep Trade Agreements database (Hofmann, Osnago, and Ruta, 2017) classifies the content of over 280 agreements into dozens of policy areas and reveals that agreements with similar headline coverage can differ dramatically in substantive provisions. The DESTA depth index (Dür, Baccini, and Elsig, 2014) provides an independent coding across seven policy areas, confirming the same heterogeneity from a different conceptual framework. Drawing on

these provision-level databases, Mattoo, Mulabdic, and Ruta (2022) revisited the classical question of whether preferential agreements create new trade or divert existing flows, finding that deep agreements generate more trade creation and less trade diversion than shallow ones, and that some deep provisions have a public-good character that increases trade even with non-members. At the firm level, deeper agreements benefit exporters unevenly across sectors: industries more exposed to non-tariff barriers gain disproportionately from provisions that address regulatory cooperation (Neri-Lainé, Orefice, and Ruta, 2023), and the distributional consequences of preferential trade liberalization are concentrated among larger and more productive firms (Baccini, Pinto, and Weymouth, 2017). Taken together, these results point to a consistent conclusion: the effects of trade agreements depend critically on *what* is inside them. A natural follow-up is whether the same holds for environmental clauses, and what effects those clauses produce on trade patterns.

The interaction between environmental regulation and trade has been the subject of a substantial theoretical literature and a growing empirical one, organized around two competing mechanisms. The first is the pollution haven hypothesis, formalized by Copeland and Taylor (1994), which predicts that stricter environmental regulations increase compliance costs for pollution-intensive producers, causing production to relocate to countries with weaker standards. The mechanism operates through the composition channel: as regulation tightens in one country, its comparative advantage in dirty goods erodes, and trade patterns adjust accordingly. The second mechanism is the green upgrading or Porter hypothesis (Porter and Linde, 1995), which predicts the opposite: well-designed environmental regulations can trigger innovation and improve firm competitiveness, ultimately more than offsetting the direct costs of compliance. Early empirical evidence on this channel shows that regulatory stringency is associated with increased patenting activity in US manufacturing industries, although with limited evidence that the resulting innovations translate into measurable competitive gains (Jaffe and Palmer, 1997). These two mechanisms are not mutually exclusive, and whether environmental provisions in trade agreements push trade patterns in one direction or the other is ultimately an empirical question.

The empirical literature on the pollution haven hypothesis has produced mixed results, in large part because of the difficulty of separating the effect of environmental regulation from other determinants of trade patterns. As Copeland and Taylor (2004) document in an extensive survey, the theoretical logic is compelling but the evidence is sensitive to how one controls for confounding factors. An influential decomposition by Antweiler, Copeland, and Taylor (2001) separates the effect of trade on pollution into three channels: a *scale* effect, whereby more trade increases total production and therefore total pollution; a *technique* effect, whereby the income gains from openness induce stricter regulation and the adoption of

cleaner technologies; and a *composition* effect, whereby trade leads countries to specialize according to comparative advantage, potentially concentrating pollution-intensive production in countries with weaker environmental standards. Using data on sulfur dioxide concentrations across 43 countries over the period 1971–1996, they find that the technique effect dominates: freer trade appears, on net, to reduce pollution, because the income gains from openness induce stricter regulation. The evidence on whether regulatory differences drive the location of pollution-intensive production is more cautious: while some effect exists, it is small relative to other determinants of industrial geography (Brunnermeier and Levinson, 2004). Levinson and Taylor (2008) take on this challenge directly by instrumenting for abatement costs. Using trade data for 130 US manufacturing industries over the period 1977–1986, they find that industries with the largest increases in pollution abatement costs experienced the largest increases in net imports, consistent with the pollution haven prediction once one addresses the econometric difficulties that limited earlier studies. The dirty-industry classification used in the present paper draws on this tradition, specifically the taxonomy introduced by Mani and Wheeler (1998) and Low and Yeats (1992), who identified a set of pollution-intensive sectors — pulp and paper, industrial chemicals, petroleum refining, iron and steel, non-ferrous metals — that have become the standard reference for empirical work on trade and pollution.

Whether environmental provisions embedded in trade agreements can produce effects analogous to those of unilateral environmental regulation has been addressed by a small but growing literature. Two recent surveys reach a similar conclusion: the evidence on the effects of environmental provisions in trade agreements remains thin, and the results are highly dependent on the type of provision and the empirical setting (Felbermayr, Peterson, and Wanner, 2025; Gutsch et al., 2025). In light of these findings, a handful of empirical studies have produced results that point in a consistent direction.

Brandi et al. (2020) address a question closely related to the one posed in this paper. Combining the TREND environmental provision database (Morin, Pauwelyn, and Hollway, 2017) with bilateral trade flows from UN Comtrade for a panel of over 180 countries, they estimate the effect of different types of environmental provisions on the composition of developing countries’ exports. Their key finding is that the effect depends critically on the type of provision: trade-restrictive provisions, which impose conditions on market access, reduce dirty exports by about 5 percent of the mean; liberal provisions, which facilitate trade in environmental goods, raise the green export share. Provisions expressed as general cooperation language, by contrast, show no measurable effect. The estimation relies on a gravity framework at the aggregate bilateral level, with country-pair and year fixed effects, and the identification comes from cross-sectional variation in the type and number of environmental

provisions across roughly 680 agreements. Using a similar gravity framework, Berger et al. (2020) find that membership in PTAs with more environmental provisions is associated with lower bilateral exports overall, with the effect concentrated in South-North trade flows and not robust to PPML estimation. The contrast between these two sets of findings is instructive. The key difference lies in how each study treats provision heterogeneity: Brandi et al. (2020) decompose environmental content into distinct categories — restrictive, liberalizing, and cooperative — and find that only provisions with a direct trade mechanism produce measurable effects. Berger et al. (2020), by contrast, use an aggregate count that combines provisions with and without such mechanisms into a single index, which may obscure the heterogeneity that drives the result in Brandi et al. (2020).

Abman, Lundberg, and Ruta (2024) exploit a different margin of variation. Rather than comparing countries with and without environmental provisions, they compare agreements that contain specific clauses on forest preservation and biodiversity protection with agreements that do not, using satellite-based deforestation data as the outcome. Their identification strategy relies on within-agreement variation — not all agreements with environmental chapters include forest-specific clauses — and they find that these clauses offset the increase in deforestation that typically follows the entry into force of a trade agreement. The finding is important because it demonstrates that environmental provisions can have real environmental effects, but only when they contain a specific and identifiable mechanism. Along similar lines, Baghdadi, Martinez-Zarzoso, and Zitouna (2013) show that PTA partners with environmental provisions experience convergence in emissions of sulfur dioxide and carbon dioxide, using a gravity-type model estimated with PPML on a panel of bilateral trade relationships over the period 1980–2005.

On the methodological side, a relevant precedent is Cherniwchan (2017), who uses within-plant US manufacturing data around the implementation of NAFTA to identify the effect of tariff reductions on emissions intensities. His finding that emissions changes operate through individual producers’ behavior rather than through market selection demonstrates the value of micro-level data for disentangling the channels through which trade policy affects environmental outcomes. However, his result concerns a *level* effect of trade liberalization on emissions — the kind of effect that the present design cannot identify, because environmental provision depth enters in force with the agreement and is absorbed by firm-destination-year fixed effects (Section 4). Shapiro (2021) documents that most countries’ tariff structures implicitly subsidize pollution-intensive sectors and tax clean ones, a systematic environmental bias that environmental chapters are meant to work against. His replication data, which include China-specific CO₂ emission intensities by industry, supply the continuous pollution-intensity measure used as a robustness check in Section 5.

A recent strand of work has brought the question specifically to China. Yue and Lin (2024) study the effect of environmental provisions in PTAs on the export of energy-consuming products from China, finding that provisions inhibit these exports. Zhu and C. Sun (2026) use the TREND database to examine whether environmental provisions in China’s trade agreements promote green transformation, and report that EPs reduce the export of dirty goods while increasing the export of green products. Zhu and C. Sun (2025) focus on a different channel, documenting that environmental provisions in PTAs are associated with improvements in the quality of Chinese firms’ exports and with increased imports of intermediate goods, consistent with a green upgrading mechanism. These studies provide suggestive evidence that environmental provisions may affect Chinese trade patterns, but they rely on identification strategies that do not fully absorb agreement-level confounders. In particular, the specifications used in these studies lack fixed effects saturated enough to separate the effect of environmental provisions from the effect of the agreement itself and from other bilateral characteristics that may be correlated with both EP content and trade outcomes. The present paper addresses this concern through a triple-difference design with firm-destination-year fixed effects, which absorb the agreement and everything constant within a firm’s relationship with a market in a given year, isolating the differential response of green and dirty products. The contrast with Zhu and C. Sun (2026), whose customs data overlap with those used here, deserves to be made explicit. Their finding that environmental provisions raise clean and reduce dirty export shares is identified from variation *across* firms and destinations — precisely the channel that firm-destination-year fixed effects absorb in the present design. The two sets of results are therefore not in contradiction: Section 4 documents that moving from the across-firm comparison (the collapsed panel) to the within-firm comparison (the full panel) shrinks the dirty coefficient by a factor of 2.7, a direct measure of how much of the aggregate association reflects between-firm composition rather than reallocation within the exporting firm — and only the latter is the margin the policy argument has in mind.

What emerges from this body of work is a consistent pattern: environmental provisions in trade agreements can have measurable effects on trade composition and environmental outcomes, but only when they contain specific, enforceable clauses with an identifiable mechanism. General cooperation language, which characterizes the vast majority of environmental chapters, does not produce detectable effects in studies that distinguish among different categories of provisions, such as restrictive, liberalizing, and cooperative clauses. This paper occupies the mirror position of Brandi et al. (2020): while they identify effects driven by the subset of agreements with specific provisions, the Chinese agreements in the period under study are dominated by exactly the kind of broad cooperation language that produced no effect in their decomposition. The present null result is therefore not a puzzle, but rather the

predicted outcome of the existing literature’s own heterogeneity findings, applied to a sample where the treatment lacks the specificity and enforceability that drive the effects documented elsewhere. What this paper adds is the granularity of the test: rather than aggregate bilateral flows, it exploits the universe of Chinese customs transactions to ask whether the same firm shifted its product mix in response to environmental provisions, a question that aggregate data cannot answer. The paper also illustrates the inference challenges that arise when treatment varies across a small number of policy units, engaging the growing literature on inference with few treated clusters (Abadie et al., 2023; Cameron, Gelbach, and Miller, 2008; Conley and Taber, 2011; MacKinnon and Webb, 2017) to show that a result that appears significant under asymptotic inference dissolves under bootstrap and permutation tests.

3 Data

The analysis combines two types of data. On the treatment side, we draw on the environmental content of China’s preferential trade agreements as coded independently by the World Bank Deep Trade Agreements database (Hofmann, Osnago, and Ruta, 2017) and by TREND (Morin, Pauwelyn, and Hollway, 2017). On the outcome side, we use the universe of Chinese customs transactions for 2000–2015 at the firm–HS6–destination–year level: 45.8 million observations after excluding Hong Kong and Macao, covering 462,651 exporting firms, 236 destination economies, and between 4,450 and 4,766 HS6 product codes depending on the year. The outcome variables are the logarithm of export value, the logarithm of export quantity, and the logarithm of unit value. Products are classified as green (OECD CLEG list), dirty (pollution-intensive sectors), or neutral, and the triple-difference design exploits composition differences across these categories within firm–destination–year cells.

3.1 Chinese PTAs and their environmental content

Between 2000 and 2015, China signed preferential agreements covering 25 destination economies, of which 23 enter the main estimation sample after excluding Hong Kong and Macao (Table 1). Three features of this treatment landscape shape everything that follows.

First, the ASEAN–China agreement covers eleven destinations, all of which share identical EP index values because they are parties to the same treaty. For identification purposes, these eleven destinations provide a single source of treatment variation rather than eleven independent ones; the total of 23 treated destinations thus corresponds to 13 distinct EP profiles—combinations of depth and entry timing—of which the ASEAN profile alone covers

Table 1: Chinese PTAs in force, 2000–2015

Agreement	Destinations	In force	WB EP	TREND EP
Bangkok Agr./APTA	Bangladesh, India, S. Korea, Laos, Sri Lanka	2002	1	1
ASEAN–China	Brunei, Cambodia, Indonesia, Malaysia, Myanmar, Philippines, Singapore, Thailand, Timor-Leste, Vietnam	2005	6	4
China–Chile	Chile	2006	5	30
China–Pakistan	Pakistan	2007	3	12
China–New Zealand	New Zealand	2008	7	53
China–Singapore	Singapore	2009	7	15
China–Peru	Peru	2010	12	47
China–Costa Rica	Costa Rica	2011	4	50
China–Iceland	Iceland	2014	6	28
China–Switzerland	Switzerland	2014	14	53
China–Australia	Australia	2015	3	24
China–S. Korea	South Korea	2015	17	72
<i>Excluded from main specification</i>				
CEPA Hong Kong	Hong Kong	2003	4	5
CEPA Macao	Macao	2003	5	5
Total treated destinations (excl. HK–MO)				23

Entry-into-force years from the World Bank Deep Trade Agreements database and TREND. WB EP = count of environmental-provision categories (World Bank); TREND EP = count of individual environmental clauses (TREND). Values shown are the maximum observed over the sample period. Three destinations experience within-country increases in EP depth: Laos (Bangkok 2002 → ASEAN 2005), Singapore (ASEAN 2005 → bilateral 2009), and South Korea (Bangkok 2002 → bilateral 2015). Laos is listed under the Bangkok Agreement, its first agreement; it is also a party to ASEAN–China from 2005, so eleven destinations are ASEAN–China parties in total. Timor-Leste is assigned to the ASEAN–China agreement in the country lists used to expand agreements into destination–year rows, although it acceded to ASEAN only in 2022. It accounts for 0.02% of observations, and excluding it moves the dirty coefficient by less than 10^{-6} (leave-one-out row `senza_144`).

nine destinations. Standard errors clustered by destination and the permutation inference described in Section 4.3 account for this structure.

Second, EP content is fixed at signature and almost never changes afterwards. Only three destinations (Laos, Singapore, and South Korea) exhibit within-country variation over time, and in each case depth rises, never falls. There is no treatment reversal in the sample, which means estimators designed for treatments that switch on and off are not needed here and would be misspecified.

Third, Hong Kong and Macao (CEPA, 2003) are entrepôt economies. They account for 24.4% of treated observations and 50.1% of treated export value, but a large fraction of what the customs data record as Chinese exports to these destinations are goods transshipped and re-exported to third countries at a markup (Feenstra and Hanson, 2004). Including them would mix the demand of the true final destination with the bilateral fixed effects designed to absorb it, since the recorded destination and the actual final market differ systematically.¹

Entry into the treated group is concentrated in two large cohorts (2002 and the 2005 ASEAN accession) and then trickles in one destination at a time in most subsequent years. Of the 236 destinations in the panel, 23 are ever treated and 213 are never treated within the sample window.

EP content is measured with two independent codings. The World Bank Deep Trade Agreements database (Hofmann, Osnago, and Ruta, 2017) yields EP_{dt}^{WB} , the count of environmental-provision categories in force with destination d at time t , and a non-environmental depth control TD_{dt} (total non-environmental provisions). TREND (Morin, Pauwelyn, and Hollway, 2017) yields EP_{dt}^{TR} , a count of individual environmental clauses, and sub-indices covering green market access, enforcement stringency, and hard versus soft provisions. Using two codings independently constructed from different frameworks guards against the idiosyncrasies of either; as shown below, they agree on the main findings.

Figure 1 traces the expansion of China’s PTA network and the evolution of EP content over the sample period. The early agreements (Bangkok 2002, ASEAN 2005) contain minimal environmental language; substantive environmental chapters appear only from New Zealand (2008) onward, with the deepest content in the Switzerland and South Korea agreements (2014–2015). Figure 2 shows the distribution of maximum EP depth across the 23 treated destinations: the WB index is concentrated at low values (1 and 6), reflecting the large Bangkok and ASEAN cohorts, while the TREND count is more dispersed.

¹Results including Hong Kong and Macao are reported in Section 5. Point estimates are substantively unchanged; the main effect is wider confidence intervals due to the additional noise from entrepôt flows.

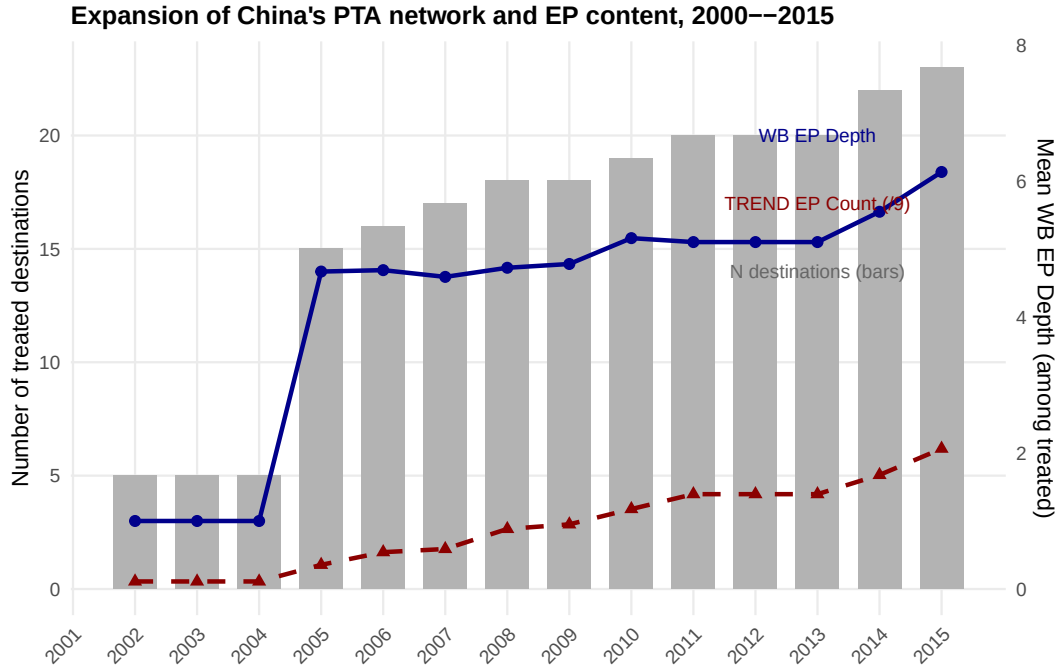


Figure 1: Expansion of China's PTA network and EP content, 2000–2015
Notes: Bars show the number of treated destinations (excl. HK–MO) with positive EP depth in each year. Lines show the mean WB EP Depth (solid) and TREND EP Count (dashed, scaled) among treated destinations.

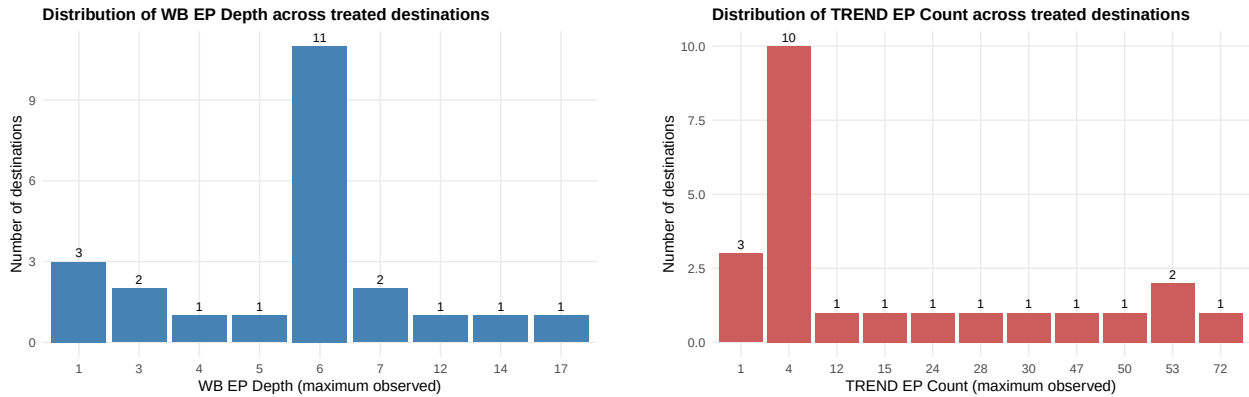


Figure 2: Distribution of EP depth across treated destinations
Notes: Each bar counts the number of treated destinations (excl. HK–MO) at a given maximum EP depth level. Left: WB EP Depth. Right: TREND EP Count.

Destinations covered by Chinese PTAs, 2000–2015

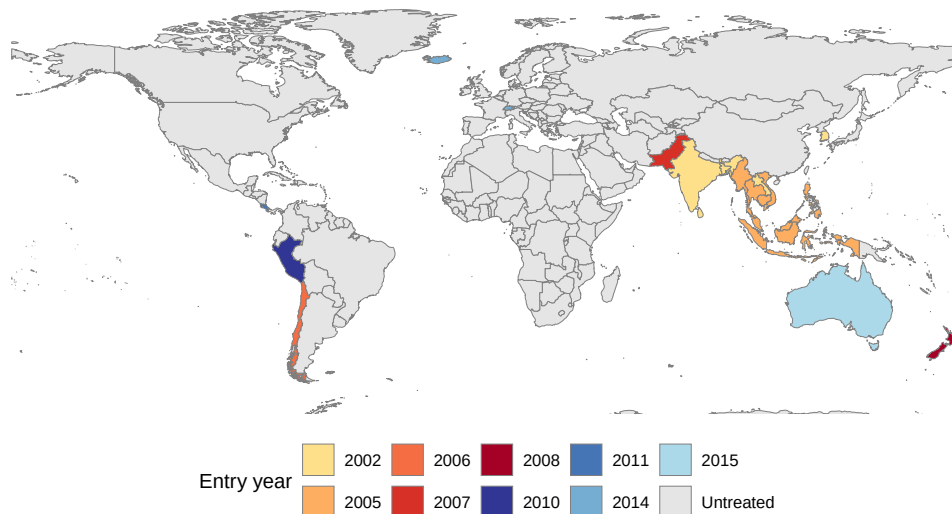


Figure 3: Geographic coverage of Chinese PTAs, 2000–2015

Notes: Colored countries are destinations covered by a Chinese PTA with environmental provisions during the sample period. Color indicates the year of entry into force. Hong Kong and Macao (excluded from the main specification) are not shown.

3.2 Customs data and product classification

The trade data are the universe of Chinese export customs transactions for 2000–2015, sourced from the General Administration of Customs of China. Transactions are aggregated to the firm–HS6–destination–year level, yielding 49.2 million observations (45.8 million once Hong Kong and Macao are excluded), 462,651 exporting firms, 236 destinations, and between 4,450 and 4,766 HS6 product codes depending on the year (the count shifts with the 2007 HS revision). Three outcome variables are used throughout: the logarithm of export value ($\ln x_{fpt}^{val}$), the logarithm of export quantity ($\ln x_{fpt}^{qua}$), and the logarithm of unit value ($\ln x_{fpt}^{uv}$, defined as value divided by quantity). Of these, export value is the primary outcome; quantity and unit value serve as a decomposition check, since any shift in value must operate through one or both margins.

Green products are the 248 HS6 codes of the OECD Combined List of Environmental Goods (CLEG), as compiled in Sauvage (2014). The customs panel uses the HS1996 vintage while the OECD list is defined in HS2012, so the list was concorded. Of the 248 codes, 246 have a one-to-one mapping to HS1996 and are translated directly. The remaining two (codes 871411 and 871419, which correspond to a single HS1996 heading split into two sub-codes in HS2012) are retained at the original HS1996 level — no information is lost, only the finer HS2012 distinction is unavailable. Export-value continuity checks around the 2007 vintage

revision find no suspicious breaks on any translated code.²

Dirty products belong to the classic pollution-intensive sectors of Mani and Wheeler (1998) and Low and Yeats (1992): pulp and paper, industrial chemicals, petroleum refining, iron and steel, and non-ferrous metals, with cement added in an extended variant.³

Green products account for 11.5% of firm-level observations (excluding Hong Kong and Macao) and dirty products for 7.0%. These shares are consistent with the underlying classification: the CLEG contains 248 out of roughly 5,000 HS6 codes (5%), but Chinese firms export green products more than proportionally relative to the number of codes; conversely, the dirty sectors cover 1,139 codes (23%) but are exported by fewer firms. The remaining neutral products form the comparison group. Of all observations in the main panel, 20.3% involve a destination under an agreement with positive EP content in that year.

Table 2 summarizes the main variables used in the analysis, and Table 3 reports their summary statistics.

Table 2: Variable descriptions

Variable	Notation	Description
Log export value	$\ln x_{fpt}^{val}$	Logarithm of export value (USD)
Log export quantity	$\ln x_{fpt}^{qua}$	Logarithm of export quantity
Log unit value	$\ln x_{fpt}^{uv}$	Log value minus log quantity
Green product	g_p	= 1 if HS6 code is in the OECD CLEG list
Dirty product	b_p	= 1 if HS6 code belongs to a pollution-intensive sector
EP depth (WB)	EP_{dt}^{WB}	Count of environmental-provision categories (World Bank)
EP count (TREND)	EP_{dt}^{TR}	Count of individual environmental clauses (TREND)
Non-env. depth	TD_{dt}	Count of non-environmental provisions (World Bank)
Log tariff	τ_{pdt}	Logarithm of (1+ applied MFN tariff rate)
Log HHI (BACI)	HHI_{pdt}	Log Herfindahl index of destination-product concentration

Subscripts: f = firm, p = HS6 product, d = destination, t = year.

Figure 4 plots the share of green and dirty products in total export value over time, separately for treated and untreated destinations. The two groups track each other closely throughout the sample period, with no visible divergence after agreements enter into force — consistent with the regression results reported below.

²As a check on the classification itself, the specification was re-estimated restricting green products to the 54 codes of the APEC Environmental Goods List (Sauvage, 2014), on which there is explicit multilateral political consensus and which form a complete subset of the OECD list. With the WB index the green coefficient flips sign to +0.0050 (s.e. 0.0127, $p = 0.69$); with TREND it moves from +0.0018 to +0.0032 ($p = 0.13$). Both remain indistinguishable from zero, with standard errors roughly double, as expected from restricting to 22% of the original green sample.

³Products are mapped from ISIC Rev.2 sector codes to HS1996 via the official WITS/UNSD HS1996–ISIC Rev.3 concordance table, yielding 1,139 HS6 codes. Seventeen codes that appear in both the green and dirty lists (materials produced by dirty industries but used for environmental purposes, such as some insulation products) are assigned to the green list; the two categories are mutually exclusive.

Table 3: Summary statistics (full panel, excl. HK-MO)

Variable	N	Mean	Median	Std. Dev.	Range
$\ln x_{fpt}^{val}$	45,781,181	9.264	9.360	2.546	[0, 23.3]
$\ln x_{fpt}^{qua}$	45,470,053	7.312	7.595	3.213	[-6.9, 25.7]
$\ln x_{fpt}^{uv}$	45,470,040	1.986	1.609	2.306	[-13.0, 20.4]
g_p (green)	45,781,211	0.115	0	0.319	{0, 1}
b_p (dirty)	45,781,211	0.070	0	0.255	{0, 1}
EP_{dt}^{WB}	45,781,211	0.966	0	2.383	[0, 17]
EP_{dt}^{TR}	45,781,211	2.051	0	8.165	[0, 72]
τ_{pdt}	41,117,073	1.658	1.792	1.027	[0, 7.3]
HHI_{pdt}	45,306,336	-1.540	-1.610	0.592	[-3.3, 0]

45,781,211 firm-HS6-destination-year observations after excluding Hong Kong and Macao. Observation counts differ across variables due to missing tariff data and quantity/unit-value availability. Mean, median, and standard deviation omitted for binary variables with more than two categories.

Table 4: Summary statistics (collapsed panel, excl. HK-MO)

Variable	N	Mean	Median	Std. Dev.	Range
$\bar{y}_{pdt}$ (mean $\ln x^{val}$)	3,773,498	8.974	9.029	1.957	[0, 20.7]
Cell count (n_{pdt})	3,773,498	12.13	3	43.64	[1, 8,946]
EP_{dt}^{WB}	3,773,498	0.669	0	1.980	[0, 17]
EP_{dt}^{TR}	3,773,498	1.438	0	6.828	[0, 72]
g_p (green)	3,773,498	0.084	0	0.278	{0, 1}
b_p (dirty)	3,773,498	0.140	0	0.347	{0, 1}

3,773,498 HS6-destination-year cells. $\bar{y}_{pdt}$ is the within-cell mean of log export value across firms. Cell count n_{pdt} is the number of firm-level observations in each cell (used as regression weight).

Table 5: Summary statistics (combined: full and collapsed panels)

Variable	Full panel			Collapsed panel		
	N	Mean	S.D.	N	Mean	S.D.
$\ln x^{val} / \bar{y}_{pdt}$	45,781,181	9.264	2.546	3,773,498	8.974	1.957
$\ln x^{qua}$	45,470,053	7.312	3.213	—	—	—
$\ln x^{uv}$	45,470,040	1.986	2.306	—	—	—
g_p (green)	45,781,211	0.115	0.319	3,773,498	0.084	0.278
b_p (dirty)	45,781,211	0.070	0.255	3,773,498	0.140	0.347
EP_{dt}^{WB}	45,781,211	0.966	2.383	3,773,498	0.669	1.980
EP_{dt}^{TR}	45,781,211	2.051	8.165	3,773,498	1.438	6.828
τ_{pdt}	41,117,073	1.658	1.027	—	—	—
HHI_{pdt}	45,306,336	-1.540	0.592	—	—	—
Cell count (n_{pdt})	—	—	—	3,773,498	12.13	43.64

Full panel: 45,781,211 firm–HS6–destination–year observations. Collapsed panel: 3,773,498 HS6–destination–year cells. Both exclude Hong Kong and Macao. The green share is higher in the full panel (11.5% vs. 8.4%) because green products are exported by more firms per product–destination cell; conversely, the dirty share is higher in the collapsed panel (14.0% vs. 7.0%) because dirty products have fewer exporting firms per cell.

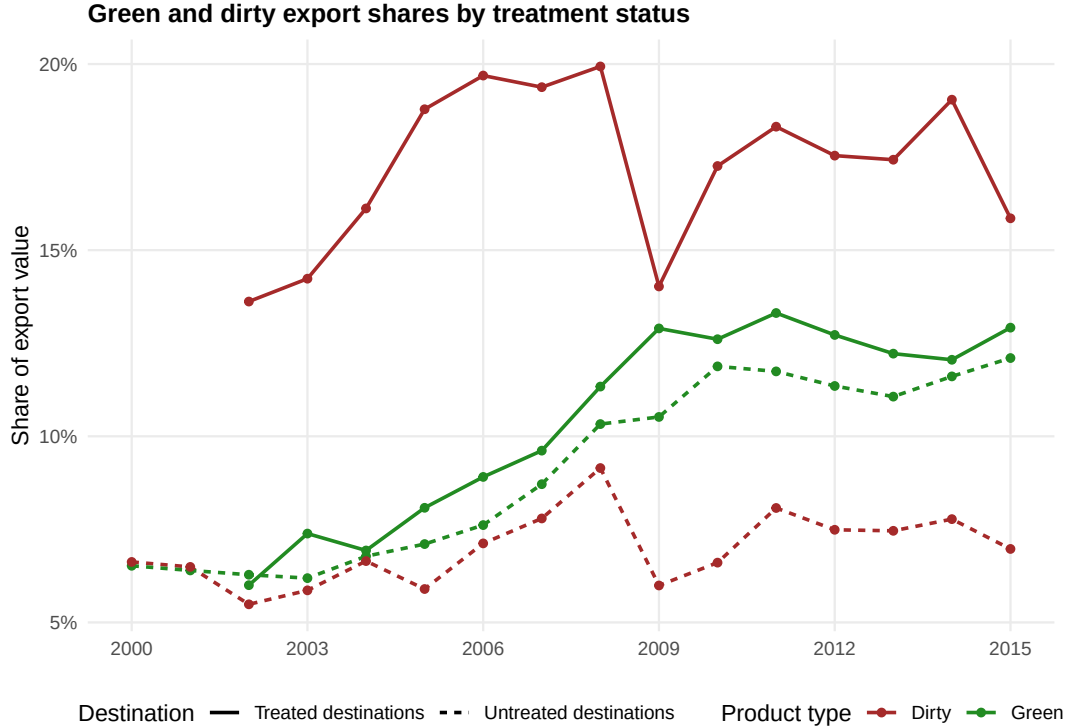


Figure 4: Green and dirty export shares by treatment status, 2000–2015
Notes: Share of export value accounted for by green and dirty products, computed separately for destinations with positive EP depth (treated) and destinations with no PTA or no EP content (untreated) in each year. Excludes Hong Kong and Macao.

3.3 Estimation samples

The analysis rests on four estimation objects built from the same underlying panel (Table 6).

Table 6: Descriptive statistics

<i>Full firm-level panel (excl. HK-MO)</i>	
Observations (firm-HS6-dest-year)	45,781,211
Exporting firms	462,651
HS6 products	4,450–4,766
Destinations	236
of which treated (EP depth > 0)	23
Distinct EP profiles (depth $\times$ timing)	13
Share of observations: green products	11.5%
Share of observations: dirty products	7.0%
Share of observations under an EP agreement	20.3%
<i>Collapsed panel (HS6-dest-year)</i>	
Cells	3,773,498
of which green / dirty	8.4% / 14.0%
Estimation sample (post-singleton)	3,681,023
<i>PPML zero-filled grid (HS6-dest-year, excl. HK-MO)</i>	
Cells	8,179,904

Hong Kong and Macao excluded throughout (they account for 24.4% of treated observations and 50.1% of treated export value). Product classification as described in the text. HS6 product count varies by year due to the 2007 HS revision.

The *full firm-level panel* is the main estimation object, with 45.8 million observations at the firm-HS6-destination-year level. The outcome variables are $\ln x_{fpt}^{val}$, $\ln x_{fpt}^{qua}$, and $\ln x_{fpt}^{uv}$; the main fixed effects are firm-product-destination (θ_{fpt}), firm-destination-year (θ_{fdt}), and product-year (θ_{pt}). Standard errors are clustered by destination.

The *collapsed panel* aggregates the firm dimension, yielding 3.77 million HS6-destination-year cells. The outcome in each cell is the mean of log exports across the underlying firm-level observations, and regressions are weighted by the cell count (the number of firm-level observations in each cell). Fixed effects are product-destination (pd), destination-year (dt), and product-year (pt). The collapsed panel serves two purposes: it is a computational workhorse for the bootstrap and permutation inference battery, and it provides an internal cross-check against the full panel.⁴

⁴With 45.8 million observations and three sets of high-dimensional fixed effects, the full panel requires substantial computational resources: a single OLS estimation takes approximately 30 minutes, and the wild cluster bootstrap (9,999 replications) runs for several hours. The collapsed panel reduces dimensionality by a factor of twelve, making the full inference battery feasible on a single machine. The equivalence between

The *zero-filled PPML grid* (8.2 million cells) is built from the full panel by first aggregating to HS6–destination–year cells and then adding product–destination pairs with no recorded trade in a given year, conditional on the pair having at least one positive flow elsewhere in the sample period. This conditional zero-fill — rather than a complete cross-join of all products, destinations, and years — keeps the grid computationally tractable while capturing the relevant extensive margin: the appearance and disappearance of trade flows within product–destination pairs that are active at some point. A zero-fill at the firm–product–destination–year level would generate billions of rows and exceed available memory; a PPML without zero-fill would miss the extensive margin entirely, since the estimator’s advantage lies precisely in the information contained in the zeros.

The *within-firm share panel* (13.3 million firm–destination–year observations) recasts the outcome as each firm’s green export share toward a given destination in a given year. This panel is used for descriptive purposes — for example, to document how green export shares evolve over time by treatment status — but not for causal inference. The reason is that the firm–destination–year fixed effects that are central to the identification strategy absorb the overall level of each cell, leaving no variation in a share outcome when a firm exports to a single destination in a given year; the triple-difference on the full panel is the appropriate design for the causal question.

Because the triple-difference compares green and dirty products to neutral ones, the credibility of the comparison group can be probed directly. Four control-group subsamples, each tightening the comparison in the direction of a specific threat, are summarized in Table 7 and described below.

Table 7: Control-group subsamples

Subsample	Threat addressed
C-prod-HS4 neutral products in the same HS4 family as a green code	Neutral products differ from green/dirty
C-overlap products exported both to treated and untreated destinations	Extrapolation outside common support
CEM destinations destinations matched on year-2000 GDP p.c., growth, MFN tariff	Selection of partner countries
Deep vs. shallow PTA partners only, split at median EP depth	The “any-agreement” contrast itself

All observation counts refer to the full firm-level panel (excl. HK–MO).

C-prod-HS4 restricts the neutral comparison group to products belonging to the same

the two implementations is verified algebraically and numerically in Section 4.

HS4 family as a green code (106 HS4 families, covering 20.5% of observations). This guards against the concern that neutral products are too dissimilar from green ones to serve as a valid counterfactual. It should be read alongside C-overlap: multi-product firms may reallocate toward core products when trade costs fall (Eckel and Neary, 2010), potentially displacing neutral products in the same HS4 family and biasing the contrast.

C-overlap keeps only HS6 products that are exported to both treated and never-treated destinations (98.5% of all HS6 codes, 21.5 million observations). This ensures that the identifying variation does not rely on extrapolation from products traded exclusively with one group of destinations. In practice the restriction is close to non-binding: it removes 314 of the 21,519,511 observations in the estimation sample, so this row is best read as a confirmation that no identifying variation comes from products traded with only one group of destinations, not as an independent robustness design.

CEM destinations applies coarsened exact matching (Iacus, King, and Porro, 2012) at the destination level, matching treated destinations to untreated ones on year-2000 GDP per capita, GDP growth, and MFN tariff levels. The resulting sample contains 19 treated and 40 control destinations (14.0 million observations) and addresses the concern that treated and untreated destinations differ systematically on dimensions correlated with export composition.

Deep vs. shallow drops all untreated destinations and splits the remaining PTA partners at the median EP depth (16 deep, 7 shallow partner countries in the baseline sample; 5.3 million observations). This design asks whether composition differs between deep and shallow environmental chapters, conditional on having an agreement at all, and addresses the concern that any observed effect reflects the agreement rather than its environmental content. The few-cluster inference battery of Section 4.3 handles the small number of clusters in this subsample.

The reading rule that applies throughout: a genuine composition effect survives every tightening of the comparison group; an artifact of the comparison moves when the comparison moves.

4 Empirical Strategy

The empirical question outlined in the introduction faces three interrelated challenges: countries do not sign trade agreements at random, environmental depth and overall agreement depth are highly correlated, and the number of independent treatment units is small. These challenges shape every aspect of the identification strategy.

A natural first step would be to estimate the level effect of environmental provisions on

trade—regressing log exports on environmental depth with fixed effects. Two arguments, one empirical and one structural, show that this approach is uninformative in this setting.

The empirical argument comes from a saturation ladder exercise (Appendix B). Across four fixed-effects structures the EP-depth coefficient is always small—between +0.0001 and +0.0046 log points per provision—and is nominally significant in exactly one of them, the structure that combines firm–product–year with product–destination effects ($p = 0.09$). It is indistinguishable from zero in the three others, including the two most saturated. A coefficient that reaches conventional significance in one fixed-effects structure out of four, with no stable sign pattern across them, is the signature of residual selection into agreements rather than of a treatment effect (Bertrand, Duflo, and Mullainathan, 2004; Goodman-Bacon, 2021). Two features of the exercise limit how much weight it can bear, and both are stated here rather than in a footnote: none of the four structures includes the firm–destination–year absorption of equation (1), and each structure induces a different iterative singleton removal, so the four estimation samples range from 22.9 to 35.6 million observations. The exercise is therefore suggestive, and the structural argument that follows is the one that carries the section.

The structural argument is independent of the empirical results. Any fixed-effects structure that seriously addresses bilateral confounders requires a destination–year intercept, which absorbs *all* variation at the (d, t) level—including environmental depth itself. With only 13 distinct EP profiles and near-zero within-destination change over time, “environmental depth” and “having a deep agreement” are empirically indistinguishable. The interaction $EP_{dt} \times g_p$ survives the destination–year intercept because the product dimension varies *within* the (d, t) cell. This is why the specification is built on composition—the differential response of green and dirty products relative to neutral ones—rather than on levels.

One consequence is that this design cannot supply evidence of first-stage “bite”—a demonstration that EP-heavy agreements shifted aggregate trade before composition is examined—because that level effect is exactly what is confounded with the agreement itself. The paper’s evidence rests entirely on the composition margin.

4.1 The triple-difference specification

The question is whether environmental depth shifts the gap between green (or dirty) and neutral log exports among treated destinations.⁵ OLS gives equal weight to each observation, so the coefficient is a per-observation average, not a per-firm or per-destination average. The

⁵In potential-outcomes terms, the target can be written as $\beta_1 = E[(y_{fpgdt}(1) - y_{fpgdt}(0)) - (y_{fpndt}(1) - y_{fpndt}(0)) \mid d, t \text{ treated}]$, where $y(1)$ and $y(0)$ denote log exports of green product g versus neutral product n under deeper and unchanged environmental content respectively, holding the agreement fixed.

main specification is

$$\ln x_{f p d t} = \beta_1 EP_{dt} \times g_p + \beta_2 EP_{dt} \times b_p + \gamma_1 TD_{dt} \times g_p + \gamma_2 TD_{dt} \times b_p + \theta_{f p d} + \theta_{f d t} + \theta_{p t} + \varepsilon_{f p d t}, \quad (1)$$

where f indexes firms, p HS6 products, d destinations, and t years; g_p and b_p are the green and dirty product indicators defined in Section 3; EP_{dt} is environmental depth and TD_{dt} non-environmental agreement depth. Environmental depth enters the regression only through its interactions with g_p and b_p , not on its own: the level effect of EP_{dt} is absorbed by $\theta_{f d t}$, and what remains is the differential response of green and dirty products relative to neutral ones.

The three sets of fixed effects absorb different sources of confounding variation, and the choice of which to include is dictated by the structure of the treatment.

The firm–product–destination fixed effects $\theta_{f p d}$ absorb time-invariant features of each firm–product–market relationship—for example, a firm’s long-standing contract with a South Korean buyer of solar panels, or an established network for exporting steel to a particular ASEAN destination (cf. the industry–country interaction logic in Rajan and Zingales, 1998).

The firm–destination–year fixed effects $\theta_{f d t}$ are what drives identification. They place a separate intercept on every firm–destination–year cell, so that anything constant within a firm’s relationship with a market in a given year drops out: the agreement dummy, overall agreement depth, destination demand shocks, and the non-random selection of destinations into agreements. For instance, if a firm opens a distribution office in South Korea in 2012 and from that point exports more of everything there, $\theta_{f d t}$ absorbs this shift entirely. Identification comes from comparing green and dirty products to neutral ones *within* the same cell as environmental content varies across destinations and over time. In the logic of Rajan and Zingales (1998), the firm–destination–year cell plays the role of “country” and the green/dirty/neutral classification plays the role of “industry”: the coefficient is identified from the interaction of cross-sectional product characteristics with variation in the policy variable across cells.

The product–year fixed effects $\theta_{p t}$ absorb global product-level shocks common to all destinations—for instance, the worldwide expansion of the solar panel market in the late 2000s, or global fluctuations in steel prices—so that these shocks do not contaminate the composition comparison.

Selection into agreements does not disappear in this design, but its scope narrows substantially. Because $\theta_{f d t}$ absorbs the level effect of the agreement, selection can contaminate β_1 only if it operates *differentially* across product categories within the same cell—that is, only if destinations that sign EP-heavy agreements were already experiencing a differential shift in green versus neutral exports before the agreement entered into force. The most plau-

sible channel for such differential contamination is a pre-existing destination-specific trend in green demand. Section 5 addresses this residual threat directly by estimating the specification with destination-specific linear trends in the green–neutral gap, and showing that the coefficient is unchanged.

The omitted category is neutral products. β_1 measures how green products move *relative to neutral ones* as environmental depth increases; β_2 measures the same contrast for dirty products. Each coefficient is already the difference from the neutral baseline. The green–dirty differential ($\beta_1 - \beta_2$) is a derived quantity, not the parameter the regression identifies, and should not be read as such.

The same World Bank non-environmental depth control is used in both the WB and the TREND specifications: TREND does not code non-environmental provisions, so no TREND-based analogue exists.

The non-environmental depth interactions ($TD_{dt} \times g_p$ and $TD_{dt} \times b_p$) separate environmental from general agreement content. TD_{dt} counts all non-environmental provisions, but not all of these are directly relevant to green or dirty trade; it is therefore an imperfect proxy for the non-environmental component of agreement depth that might independently affect export composition. An imperfectly measured control under-controls: β_1 absorbs whatever share of general-depth variation the proxy fails to capture.⁶ Rather than assert a direction, Section 5 reports the green coefficient under four alternative depth controls and shows that the point estimate moves within a band narrower than one standard error (Table 12).

The collinearity between environmental depth and non-environmental depth deserves to be stated plainly. Across the 223 treated destination–years in the estimation sample, WB environmental depth and non-environmental depth correlate at 0.91.⁷ The TREND count is less collinear with non-environmental depth ($\rho = 0.50$), which is one reason the WB confidence intervals are wider. That two codings with such different degrees of overlap return the same null is reassuring, since a result driven by the inability to separate environmental from general depth would be expected to differ between them.

Standard errors are clustered by destination. Although environmental depth varies formally at the destination–year level, it enters in force with the agreement and afterwards

⁶The direction of the bias follows from an omitted variable argument. Decompose the true non-environmental depth relevant to composition as TD_{dt}^* and write $TD_{dt} = TD_{dt}^* + \eta_{dt}$. Then $\hat{\beta}_1 \approx \beta_1 + \delta \cdot \gamma$, where $\delta = \text{Cov}(EP_{dt}, \eta_{dt} \mid \text{controls}) / \text{Var}(EP_{dt} \mid \text{controls})$ and γ is the effect of the omitted component on composition. Since EP_{dt} and TD_{dt} are positively correlated ($\rho = 0.91$), $\delta > 0$ is plausible, and $\hat{\beta}_1$ is biased upward. With $\hat{\beta}_1 \approx 0$, the true effect is zero or negative.

⁷The Variance Inflation Factor (VIF) measures how much collinearity between regressors inflates the variance of the estimated coefficient relative to an orthogonal design. The VIF for EP depth in this specification is 5.8, somewhat above the maximum of 4.6 that Brandi et al. (2020) report for their provision-type variables on a panel of 680 agreements.

changes in only three destinations, so the treatment is effectively assigned at the destination level. Clustering at the destination–year level would create hundreds of clusters with identical treatment status within each destination, understating the dependence across observations from the same country and producing artificially narrow confidence intervals. Clustering at the destination level is more conservative: it treats all of a country’s observations as a single cluster, which is the appropriate level when the treatment is, in practice, a destination-level assignment (Abadie et al., 2023).

A further identification concern works in a conservative direction. If deeper environmental chapters tend to be signed where green demand is already rising, EP_{dt} is positively correlated with an omitted destination-specific trend, which would bias β_1 toward a spurious positive. Because β_1 is an imprecise zero, this channel, if present, implies the true effect is zero or negative—the null is a conservative rather than a lenient reading of the data.

Because environmental depth is a continuous dose switching on at staggered dates, the coefficient is a weighted average of dose- and cohort-specific composition responses, not a simple average treatment effect. With a null result, this distinction is immaterial: a weighted average of near-zero effects stays near zero regardless of the weighting scheme.⁸

4.2 The two panel implementations

The full firm-level panel is estimated after iterative singleton removal (Correia, 2017), which eliminates 24.3 million observations—predominantly small, one-off flows that contribute no identifying variation—leaving 21.5 million. Within this estimation sample, the composition contrast is identified in treated firm–destination–year cells that contain both a green (or dirty) and a neutral product: 26% of treated cells for the green margin and 12% for the dirty.

The collapsed panel, described in Section 3, aggregates to HS6–destination–year cells weighted by the number of underlying firm-level observations. The weights ensure that the weighted cell-level regression is algebraically identical to the unweighted micro regression under the same fixed effects.

The two implementations agree on the green margin (−0.0046 collapsed, −0.0023 full panel; both indistinguishable from zero). On the dirty margin they differ by a factor of roughly 2.7 (−0.0119 collapsed versus −0.0044 full panel). The gap reflects a substantive

⁸Under continuous treatment entering at staggered dates, the TWFE estimator produces a weighted average of dose- and cohort-specific effects whose weights need not be convex (Callaway, Goodman-Bacon, and Sant’Anna, 2024; Chaisemartin and D’Haultfoeulle, 2020; Goodman-Bacon, 2021). Two features limit what this costs in this setting. First, a weighted average of near-zero effects stays near zero unless the underlying effects are large and of opposite signs, of which the sub-index and leave-one-out evidence gives no indication. Second, the permutation test is agnostic about weighting: it asks whether the observed coefficient—whatever weighted average it represents—is unusual relative to placebos.

difference, not an error: the collapsed panel compares products across all firms exporting to a given destination–year, so any shift in *which* firms export dirty goods enters the coefficient. The firm–destination–year fixed effects in the full panel remove this between-firm channel and isolate reallocation *within* the firm. Approximately three-fifths of the collapsed dirty coefficient reflects between-firm composition rather than within-firm response. The full panel is therefore the reference specification—it answers whether individual firms shifted their product mix—while the collapsed panel serves as the computational workhorse for the inference battery.

The difference in the number of destination clusters between the two panels—225 in the full panel, 228 in the collapsed—is a consequence of singleton removal. With the high-dimensional fixed effects of the full panel (θ_{fpd} , θ_{fdt} , θ_{pt}), iterative removal drops all observations from 11 of the 236 destinations; the collapsed panel’s lower-dimensional fixed effects (pd , dt , pt) are less demanding and drop only 8.

4.3 Inference with few treated clusters

Of the destination clusters in the estimation sample, only 23 are treated, corresponding to 13 distinct EP profiles. With so few treated clusters, asymptotic cluster-robust standard errors are unreliable: the finite-sample approximation to the t -distribution is too optimistic, and p -values systematically over-reject the null (Cameron, Gelbach, and Miller, 2008; MacKinnon and Webb, 2017). Every headline estimate is therefore subjected to two additional tests.

Wild cluster bootstrap. The wild cluster bootstrap (Cameron, Gelbach, and Miller, 2008; Roodman et al., 2019) constructs an empirical null distribution by randomly assigning ± 1 signs to cluster-level residuals across $B = 9,999$ replications, rather than relying on asymptotic critical values. In the full panel, the bootstrap runs natively after `reghdfe` and is the version reported for the headline confidence intervals. In the collapsed panel, the fixed effects are partialled out of the data once using the Frisch–Waugh theorem—each variable is demeaned by the fixed-effects structure, and the bootstrap then operates on the resulting low-dimensional regression with only four regressors. The point estimates are identical to those from the full regression; what changes is only the computational burden, which drops from millions of fixed-effects parameters to a handful of regressors.⁹

Permutation inference. The permutation test (Fisher, 1935) asks whether the observed coefficient is unusual relative to placebos obtained by randomly reassigning EP profiles (depth

⁹Two approximations apply to the collapsed-panel bootstrap only. First, the bootstrap treats the product–year fixed effects as if nested within the destination cluster (they are not; product–destination and destination–year are). Second, the small-sample corrections see the four residualized regressors rather than the four plus absorbed fixed effects. Neither approximation applies to the full-panel bootstrap.

and timing, jointly) across treated destinations, with 1,000 replications. The test operates only among treated destinations because the question it answers is whether the *specific EP content*—not the agreement itself—matters for composition. Permuting between treated and untreated destinations would test a different hypothesis (the effect of the agreement as such), which the level-effect analysis has already shown to be unidentifiable.

The null hypothesis is that, given the set of 23 treated destinations, it does not matter which destination received which EP profile: the test permutes profiles among the treated, without adding or removing destinations from the treated group. Because depth and timing are permuted jointly, the test cannot separate the effect of environmental content from the effect of entry timing.

A feature of the permutation distribution deserves explanation. Of the 23 treated destinations, eleven are parties to the ASEAN–China agreement, and nine of them have that agreement as their only source of EP content throughout the sample: Laos entered through the Bangkok Agreement first, and Singapore signed a bilateral agreement in 2009. Swapping any two of those nine produces the same data, so the 23 treated destinations carry only 13 distinct EP profiles. The number of genuinely different permutations is therefore much smaller than the 1,000 draws, and the resulting distribution is discrete rather than smooth. This is why two implementations drawing different random permutations report somewhat different p -values ($p = 0.235$ in one versus $p = 0.278$ in the other for the dirty margin): both reflect Monte Carlo noise over a small set of distinct permutations, and both are far above any conventional significance threshold.¹⁰

5 Results

5.1 The green margin: a bounded null

Table 8 reports the main estimates. In the full firm-level panel, the $\text{EP} \times \text{green}$ coefficient is -0.0023 (s.e. 0.0039) with the WB index and -0.0001 (s.e. 0.0010) with TREND. The joint

¹⁰Estimates on the collapsed and destination-level panels were produced twice, in STATA (`reghdfe`, Correia, 2017; `boottest`, Roodman et al., 2019; `ppmlhdfe`, Correia, Guimarães, and Zylkin, 2020; `eventstudyinteract`, L. Sun and Abraham, 2021) and in R (`fixest`, `fwildclusterboot`); across 44 result files the two implementations agree on every point estimate to at least eight significant digits. Estimates on the 45.8-million-row firm-level panel exist only in STATA: `fixest` cannot complete them on the available hardware. They are validated instead by an algebraic identity—the firm-level panel estimated with $pd+dt+pt$ fixed effects reproduces the weighted collapsed-panel estimate to nine significant digits (Appendix A)—which checks the data construction of the two pipelines against each other. All reported numbers are the STATA ones. Two quantities legitimately differ between implementations and are discussed where they arise: bootstrap and permutation p -values, which carry Monte Carlo error, and the Sun–Abraham standard errors of Appendix C, which reflect a genuine methodological difference between `eventstudyinteract` and `fixest::sunab` (see Appendix C).

F-test on all four composition interactions is $p = 0.31$ (WB) and $p = 0.71$ (TREND). In the collapsed panel the WB estimate is -0.0046 ; wild cluster bootstrap $p = 0.65$; permutation $p = 0.60$. The observed green coefficient sits comfortably in the middle of the placebo distribution.

To calibrate magnitudes, consider the closest benchmark. Brandi et al. (2020) find that one liberal environmental provision raises the green share of developing-country exports by around 0.4 percentage points — roughly $+0.16$ log points in the metric of equation (1) — per provision. Section 4.3 argued that asymptotic inference is unreliable with 23 treated clusters; that argument extends to interval estimates, so the bounds cited here are wild cluster bootstrap intervals. The full-panel 95% bootstrap confidence interval for $EP \times green$ is $[-0.0353, +0.0355]$ per WB provision (collapsed: $[-0.0182, +0.0317]$). The corresponding asymptotic interval, $[-0.0100, +0.0055]$, is roughly six times narrower — reporting that level of precision would claim the kind of certainty this paper’s own inference section declares unwarranted.

On the bootstrap bound, the upper limit is about a quarter of the Brandi equivalent effect: the design rules out their point estimate and anything above roughly a quarter of it, and is uninformative below that threshold. This is a weaker statement than asymptotic inference would support, and it is the one the design can sustain. In per-standard-deviation terms, a one-SD increase in WB EP depth (approximately 2.383 provisions, inclusive of never-treated destination-years at $EP = 0$) moves green relative to neutral exports by at most -8.4% to $+8.5\%$ at the bootstrap bounds.

5.2 Stability across control groups

Across nine designs — the four control-group subsamples, the two baseline specifications, and three full-panel variations — the $EP \times green$ coefficient stays between -0.0002 and -0.0046 and is never significant (Table 10). The stability is the result: an artifact of the comparison group would move when the comparison group changes.

5.3 Dynamics

Figure 5 plots the event study of the green and dirty versus neutral gap around entry into force, using never-treated destinations as controls. Differential pre-trends are flat for both margins. Before taking that as reassurance, a clarification is in order: parallel trends is a claim about the unobserved counterfactual path of treated destinations *after* treatment. A flat pre-trend rules out the most easily detectable violations — a composition gap already in motion before the agreement — but does not establish that the assumption holds exactly

Table 8: Triple-difference estimates: EP content and export composition

	WB EP depth		TREND EP count	
	(1)	(2)	(3)	(4)
	×green	×dirty	×green	×dirty
<i>Panel A. Full firm-level panel</i>				
EP interaction	−0.0023 (0.0039)	−0.0044 (0.0022)	−0.0001 (0.0010)	−0.0009 (0.0006)
Asymptotic p	0.57	0.052	0.92	0.15
Bootstrap p	0.69	0.18	0.93	0.18
Firm–product–dest FE	Yes	Yes	Yes	Yes
Firm–dest–year FE	Yes	Yes	Yes	Yes
Product–year FE	Yes	Yes	Yes	Yes
Observations	21,519,511			
Clusters	225 destinations			
Joint F (p -value)	0.31		0.71	
<i>Panel B. Collapsed panel (HS6–dest–year cells)</i>				
EP interaction	−0.0046 (0.0070)	−0.0119*** (0.0030)	+0.0018 (0.0018)	+0.0004 (0.0016)
Asymptotic p	0.51	<0.001	0.32	0.83
Bootstrap p	0.65	0.07	0.39	0.86
Permutation p	0.60	0.28	0.16	0.82
Product–dest FE	Yes	Yes	Yes	Yes
Dest–year FE	Yes	Yes	Yes	Yes
Product–year FE	Yes	Yes	Yes	Yes
Cells	3,681,023			
Clusters	228 destinations			

All specifications include TotalDepth×green and TotalDepth×dirty controls (not shown). Wild cluster bootstrap: $B = 9,999$ (Cameron, Gelbach, and Miller, 2008; Roodman et al., 2019). Permutation: 1,000 reassignments of entire EP profiles across treated destinations. Bootstrap 95% CIs — Panel A (1)–(4): $[-0.035, +0.036]$, $[-0.043, +0.011]$, $[-0.0037, +0.0025]$, $[-0.0027, +0.0007]$; Panel B: $[-0.018, +0.032]$, $[-0.018, +0.002]$, $[-0.0017, +0.0071]$, $[-0.0031, +0.0056]$. *** $p < 0.001$ (asymptotic only; does not survive bootstrap or permutation — see Section 5.4).

Table 9: Benchmark against Brandi et al. (2020): equivalent effect and comparison with the confidence intervals

Margin	Brandi effect (log points)	Asy. CI upper	WCB CI upper/coef.	Ratio
Green (liberal)	0.1570	0.0055	0.0355	1/4 (WCB)
Dirty (restrictive)	−0.0513	−0.0044 (point est.)	—	1/12

Table 10: Stability of the EP×green coefficient across estimation designs (WB index)

Sample / design	Obs.	EP×green	Asymptotic p
Full panel (firm level)	21.5M	−0.0023	0.57
Collapsed panel	3.7M cells	−0.0046	0.51
<i>Control-group subsamples</i>			
C-prod-HS4 (within-HS4 neutral products)	3.8M	−0.0009	0.86
CEM-matched destinations	13.7M	−0.0023	0.54
Deep vs. shallow (PTA partners only)	5.3M	−0.0022	0.53
Common support (C-overlap)	21.5M	−0.0022	0.57
<i>Full-panel variations</i>			
With controls	19.3M	−0.0002	0.94
Excluding ASEAN	19.2M	−0.0025	0.48
Including HK–Macao	23.6M	−0.0009	0.80
Fixed effects as in baseline		Yes (all rows)	
WB index		Yes (all rows)	

Each row estimates equation (1) on the indicated subsample or design; asymptotic destination-clustered p -values. Fixed effects: $fpd + fdt + pt$ (full panel); $pd + dt + pt$ (collapsed). CEM matches on log GDP p.c., growth, and MFN tariff (year 2000). Deep/shallow split at the median of maximum EP depth: 16 deep vs. 9 shallow partner destinations. TREND analogues (where estimated): full panel −0.0001 ($p = 0.91$), common support −0.0001 ($p = 0.91$), deep vs. shallow −0.0004 ($p = 0.72$). Controls: MFN tariff, HHI, antidumping indicator.

thereafter. The exercise is a falsification, not a proof.

A mild negative drift for green products in the late post-period (bins $\geq +5$) is identified only by early cohorts (ASEAN 2005, Chile 2006, Pakistan 2007, New Zealand 2008). It disappears under the Sun–Abraham estimator applied to the destination-level composition gap: the aggregated ATT is −0.042 ($p = 0.27$) for the green gap and +0.073 ($p = 0.28$) for the dirty gap — consistent with early-cohort heterogeneity rather than any effect of treatment (L. Sun and Abraham, 2021). The Sun–Abraham estimator answers a narrower question than equation (1) — it binarizes treatment (EP present or absent, discarding depth), carries no depth control, and operates on a destination–year composition gap collapsed across products — so it should not be read as a replication. It is a diagnostic for staggered-timing and cohort heterogeneity, and that is how it is used here.

The full interaction-weighted event study is reported in Appendix C. With the standard errors produced by `eventstudyinteract` — which, unlike `fixest::sunab`, accounts for the uncertainty in the estimated cohort shares used as weights — no coefficient in the $[-10, +8]$ window is individually distinguishable from zero on either margin. Two or three nominally significant coefficients appear among 28 estimates, which is what sampling noise produces under exactly flat pre-trends with 23 treated clusters. The ATT is null across all three

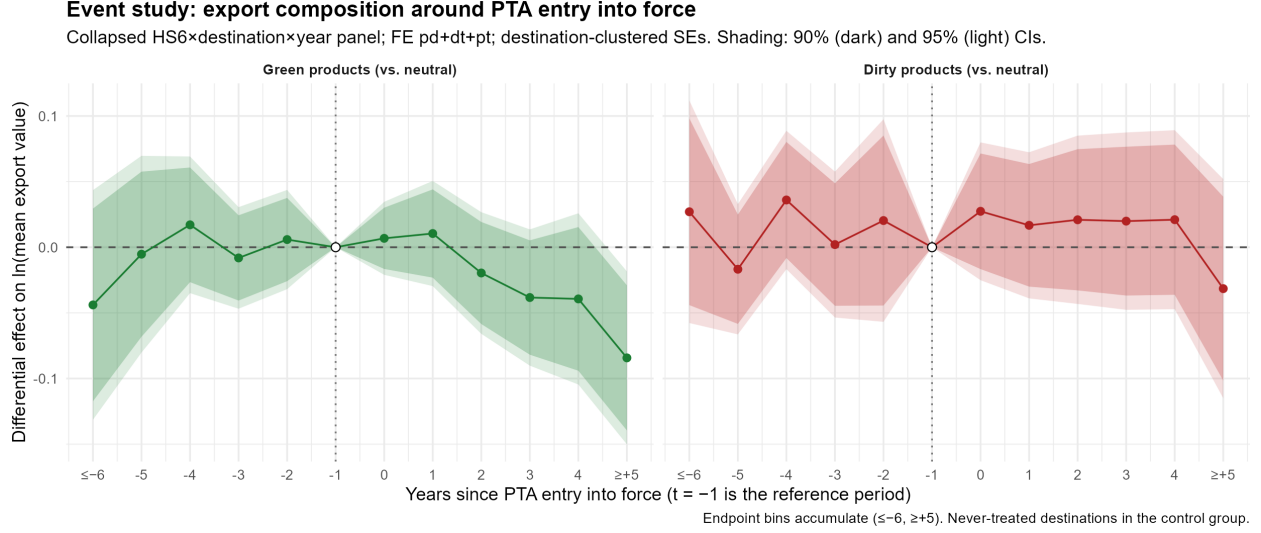


Figure 5: Event study: green and dirty vs. neutral products around PTA entry into force. Collapsed panel; FE $pd + dt + pt$; destination-clustered 95% confidence bands; $t = -1$ is the reference period; endpoint bins accumulate.

specification variants: $p = 0.28$ in the baseline, $p = 0.11$ in the $[-6, +5]$ window, $p = 0.30$ excluding the 2014–2015 cohorts.

5.4 The dirty margin: anatomy of a false positive

The $WB \times \text{dirty}$ coefficient in the collapsed panel (-0.0119 , asymptotic $p < 0.001$) warrants a subsection of its own. It illustrates, step by step, how a coefficient that looks robust under standard inference falls apart under scrutiny.

The wild cluster bootstrap raises the p -value to 0.07. Permutation inference on the exact estimating equation goes further: reshuffling entire EP profiles across the 23 treated destinations, 27.7% of placebo draws produce a coefficient as large in absolute value, giving $p = 0.28$. A coarser permutation on a destination–year–product-type aggregate reverses the sign entirely ($+0.005$, $p = 0.49$). The aggregation removes the within-cell heterogeneity across firms and product varieties that the disaggregated coefficient exploits, so the two tests are not answering exactly the same question, but they reach the same verdict. Sign instability under aggregation is itself evidence of fragility.

A leave-one-out exercise across the 23 treated destinations reveals the mechanism, and it is not the one the phrase “driven by a single country” usually implies. The *point estimate* is remarkably stable: it sits between -0.0097 and -0.0133 in every one of the 23 leave-one-out samples. Dropping large treated destinations such as India or Pakistan moves it by 1 to 4% while leaving the standard error essentially unchanged. Those are not the pivotal

observations.

What is not stable is the *precision*. Dropping Australia moves the coefficient by 13% (to -0.0103) but nearly triples its standard error, from 0.0030 to 0.0087: it is this, not the movement of the estimate, that carries the p -value to 0.24. Dropping South Korea doubles the standard error, giving -0.0097 with $p = 0.09$. Australia and South Korea are therefore not outliers exerting leverage on the estimate. They supply the variation that identifies it, and removing them leaves the design without enough information to speak — rather than with a different answer. The right description is that the coefficient is identified by a thin slice of variation concentrated in two destinations, not that those destinations pull the estimate in the wrong direction.

The exercise with the DESTA depth control makes the point sharper. With TotalDepth, Australia is the pivotal destination; with DESTA, Australia’s exclusion leaves the coefficient at -0.0110 with $p = 0.001$, and it is South Korea whose removal triples the standard error and gives $p = 0.14$. Which destination the precision depends on is itself a function of a modelling choice that has nothing to do with environmental content. A genuine identified effect does not behave this way.

The TREND index never shows the effect (bootstrap $p = 0.86$). In the full panel the coefficient is -0.0044 (asymptotic $p = 0.052$); the wild cluster bootstrap raises it to $p = 0.18$, with a 95% interval of $[-0.043, +0.011]$. The marginal signal under asymptotics is erased by robust inference on both panels independently. The honest reading is that the dirty margin is a false positive of exactly the type few-treated-cluster designs can manufacture: an estimate that is overwhelming under asymptotic standard errors and that clears none of the same battery applied to the green margin. It belongs in the paper as a descriptive pattern and a methodological warning.

5.5 Provision bundling and the limits of heterogeneity analysis

Aggregate EP indices combine provisions with an identifiable trade mechanism and cooperation language that has none. On any aggregate null, this is the natural objection — and it is correct. The response here is not to refute it but to show how extreme the bundling is, and what that implies for interpretation.

Summing WB_EP_Depth across the 25 treated destinations gives 150 provisions checked in total; 5.3% fall in `GreenLiberalization` or `StandardsNonRegression`, the two sub-indices with a direct trade mechanism. TREND, an independently constructed and far more granular coding, reaches the same conclusion: of 437 provisions across the same destinations, `GreenMarketAccess` is 0.92%. At the agreement level, TREND’s binding-obligations clause (`X5.01.01`) appears in exactly one of the 14–15 Chinese agreements in the database

(China–Korea, 2015). Table 11 puts both codings side by side.

Table 11: Trade-mechanism content as a share of total checked provisions

	Total checked	Trade-mechanism provisions	Share
WB (<code>GreenLiberalization</code> + <code>StandardsNonRegression</code>)	150	8	5.3%
TREND (<code>GreenMarketAccess</code>)	437	4	0.92%
	Agreements	With binding clause	
TREND (<code>X5.01.01</code> Binding obligations)	14–15	1 (China–Korea, 2015)	6.7%

“Total checked” sums the relevant depth index across the 25 treated destinations at each destination’s maximum (post-ent) value. Source: `Data/Merged/Merged_TREND_WB_Indices_Only.csv`.

The structural consequence is stark. Across the 223 treated destination–years in the estimation sample, the two WB sub-indices with a direct mechanism are *perfectly* collinear ($\rho = 1.000$) and non-zero in only three country–years: Korea from 2015 and Switzerland from 2014, always in the same 1:3 proportion. Their interaction terms therefore load identically in the regression (identical t -statistics; coefficients in exact 3:1 ratio), and the VIF is unbounded. Contrast this with Brandi et al. (2020), who report a maximum VIF of 4.6 across provision-type variables on their 680-agreement panel: enough agreements with enough variation in provision type to separately identify the mechanism sub-components. With 13 distinct EP profiles, this design cannot.

A reading rule applies to everything that follows: each sub-index replaces the aggregate index one at a time and is estimated on its own. The resulting coefficients are not partial effects holding other clause types constant; they absorb whatever variation in correlated sub-indices moves with each one. This is the only feasible approach, but it means the table is a set of separate comparisons, not an additive decomposition of the aggregate index.

What can be said: the mechanism-bearing bundle shows the same pattern as the aggregate. Green never responds (WB `GreenLiberalization`×green: $p = 0.90$; TREND `GreenMarketAccess`×green: $p = 0.38$). The dirty interaction is marginally negative on asymptotic p -values ($p = 0.04$ – 0.07), but this is the same asymptotics-only signal documented in Section 5.4, which does not survive robust inference.

The placebo sub-indices (without a trade mechanism) are more interesting and worth reporting carefully. TREND soft provisions show nothing on either margin (×green: $p = 0.27$; ×dirty: $p = 0.86$), consistent with the logic of the null. Regulatory space clauses do not: they carry +0.024 on green and +0.023 on dirty, and both survive the wild cluster bootstrap ($p = 0.046$ and $p = 0.022$, $B = 9,999$). This is the only place in the paper where a sub-index without a trade mechanism registers a signal that robust inference sustains.

Two features bound what can be read into it, and together they leave the conclusion weak.

First, the two margins move in tandem: the green–dirty differential is $+0.0017$ with $p = 0.80$, meaning the coefficient describes green and dirty exports rising by the same amount against the neutral baseline — not a shift in composition toward cleaner goods. Second, and more fundamentally, the regulatory-space sub-index is not a clean placebo. It accounts for 71.5% of the TREND count across treated destination–years and correlates 0.90 with TotalDepth; in this specification the depth control flips to significantly negative ($\times\text{green}$: -0.0011 , $p = 0.006$). Two regressors correlated at 0.90 splitting into a large positive and a large negative is the same bundling problem applied to the placebo, not a cleanly identified channel. The defensible conclusion is the weaker one: this design cannot separate a regulatory-space effect from general agreement depth, so neither the positive signal nor its absence can serve as a clean falsification test.

Enforcement provisions (dispute-settlement mechanisms attached to the environmental chapter) are null on both margins and both codings: WB $\times\text{green}$ $p = 0.91$, $\times\text{dirty}$ $p = 0.90$; TREND $\times\text{green}$ $p = 0.78$, $\times\text{dirty}$ $p = 0.71$.

This bundling result reconciles the null with Brandi et al. (2020). Their finding is not that environmental provisions generically matter; it is driven by the subset of agreements with explicitly liberalizing or restrictive clauses, identified separately because 680 agreements supply enough cross-agreement variation in provision type. The Chinese PTAs of this period are, on the same coding, overwhelmingly cooperation-only: mechanism-bearing content appears in two of the fourteen agreements and is perfectly collinear between its two components in-sample. A null on the aggregate index is what Brandi et al.’s own heterogeneity would predict for a sample of this composition. The two results are complementary. A second compounding difference concerns aggregation: Brandi et al. (2020) work with exporter–importer–year flows, where a composition effect can arise from the entry or exit of firms or products. The specification here saturates firm $\times$ destination $\times$ year fixed effects and identifies from reallocation *within* the same firm across products — a strictly more demanding test of the same hypothesis.

5.6 Extensive margin

Log-linear estimates condition on positive flows. If EPs create new green trade at the extensive margin, intensive estimates miss it (Santos Silva and Tenreiro, 2006). On a zero-filled HS6–destination–year grid (8.2 million cells, of which 7.9 million enter the estimation after `ppmlhdfc` drops separated and singleton observations), PPML estimates confirm the null: EP $\times$ green is $+0.0015$ ($p = 0.74$, WB) and $+0.0001$ ($p = 0.95$, TREND); EP $\times$ dirty is -0.030 ($p = 0.06$, WB) and $+0.003$ ($p = 0.55$, TREND). No green trade creation at the extensive margin; the dirty coefficient is at most marginally significant and consistent with the

non-result reading of Section 5.4.

One limitation bears noting: this grid detects green products newly exported to a destination, but not a new firm beginning to export a product that other Chinese firms already ship there, which is a common form of firm-level entry. The extensive margin is probed, not exhausted.

5.7 Within-firm share: a descriptive complement

On the firm–destination–year panel of green export shares (13.3 million observations; FE: firm–destination and year; destination-clustered), the coefficient on WB EP depth is -0.0001 ($p = 0.50$) and on TREND -0.00006 ($p = 0.043$, asymptotic). Both are economically negligible: a one-standard-deviation increase in the TREND count moves the within-firm green share by three hundredths of a percentage point, and the marginal asymptotic significance carries the same few-cluster caveat as the rest.

This module is reported as a descriptive complement, not identifying evidence. The specification regresses the green share on the *level* of EP depth with firm–destination and year fixed effects — it does not include firm–destination–year absorption and therefore inherits the collinearity with the agreement that Section 4 argues is fatal to any level specification in this setting. A null from an unidentified specification says nothing in either direction. What the module establishes, and what is worth having, is that the within-firm green share is flat in the raw conditional means, which is at least consistent with the composition null.

5.8 How much the depth control matters

Environmental depth and general agreement depth correlate at 0.96 within destination and year. Table 12 varies the depth control across four alternatives: none, aggregate TotalDepth (17 areas), a version restricted to the 14 non-environmental areas most correlated with EP content, and the independently coded DESTA index (Dür, Baccini, and Elsig, 2014).

The point estimate moves within a band of 0.0024 log points — narrower than a single standard error — stays negative throughout, and every interval contains zero. Dropping the depth control altogether moves the coefficient away from zero rather than toward it. The green null does not rest on the choice of depth control.

5.9 Environmental share of agreement content

The main specification estimates the marginal effect of one additional provision. Abman, Lundberg, and Ruta (2024) pose a different question: given that an agreement exists, does a

Table 12: EP×green under alternative depth controls (collapsed panel, WB index)

Depth control	EP×green	Asymptotic 95% CI	<i>p</i>
None	−0.0057	[−0.0118, +0.0003]	0.06
TotalDepth (17 areas) [<i>baseline</i>]	−0.0046	[−0.0182, +0.0091]	0.51
TotalDepth targeted (14 areas)	−0.0033	[−0.0188, +0.0123]	0.64
DESTA index	−0.0043	[−0.0129, +0.0042]	0.30
Product–dest FE	Yes		
Dest–year FE	Yes		
Product–year FE	Yes		
Cells	3,681,023		
Clusters	228 destinations		

Collapsed panel, weighted by cell size. Asymptotic intervals. The targeted control drops three WB areas with zero variance in Chinese agreements (labour-market regulation, visa and asylum, subsidies). DESTA covers seven policy areas coded independently of the World Bank instrument and excludes environmental provisions by construction, reducing the VIF from 5.7 to 1.9 and leaving the coefficient essentially unchanged.

more environmentally weighted agreement move composition? That framing requires restricting to partner destinations and replacing EP depth with its share of total depth, EP_{dt}/TD_{dt} . On the collapsed panel restricted to the 23 partner destinations (534,846 cells, 516,684 surviving singleton removal; the share takes 12 distinct values in [0.012, 0.068]; coefficient of variation 0.19 versus 0.62 for the level), the green interaction is −2.25 (s.e. 1.15, $p = 0.06$) and the dirty interaction −1.60 (s.e. 1.55, $p = 0.32$). The share varies far less than the level, so the standard errors are correspondingly wide; the green coefficient clears neither conventional threshold. It is reported because the contrast is the one used by the closest antecedent in the literature, and because a marginally negative coefficient is the opposite of what green trade creation would predict — not because this design identifies it well.

5.10 Sample robustness

Table 13 collects the full-panel variations. Adding controls (MFN tariff, market concentration, antidumping exposure) moves the green coefficient to an exact zero (−0.0002, $p = 0.94$). Excluding the eleven ASEAN destinations — the largest single cohort — leaves it at −0.0025 ($p = 0.48$). Re-including Hong Kong and Macao gives −0.0009 ($p = 0.80$). Across every variation, the dirty coefficient sits between −0.0040 and −0.0060 with asymptotic p -values of 0.02–0.09: persistently marginal under asymptotics, robust to none of the same battery applied to the green margin.

On the tariff control: the variable is the applied duty recorded in the customs data, not a

Table 13: Sample robustness (full firm-level panel, WB index)

Sample variation	Obs.	EP×green		EP×dirty	
		Coeff.	<i>p</i>	Coeff.	<i>p</i>
Baseline	21.5M	−0.0023	0.57	−0.0044	0.052
With controls	19.3M	−0.0002	0.94	−0.0060	0.022
Excluding ASEAN	19.2M	−0.0025	0.48	−0.0044	0.051
Including HK–Macao	23.6M	−0.0009	0.80	−0.0041	0.094
Common support	21.5M	−0.0022	0.57	−0.0044	0.052
PPML, zero-filled grid	7.9M	+0.0015	0.74	−0.0301	0.06
Within-firm green share	13.3M	−0.0001	0.50	—	
Firm–product–dest FE		Yes (all rows)			
Firm–dest–year FE		Yes (all rows)			
Product–year FE		Yes (all rows)			

WB index throughout. Asymptotic destination-clustered *p*-values; the dirty column’s marginal significance does not survive wild cluster bootstrap, permutation, or leave-one-out inference (Section 5.4). Controls: $\ln(1 + \text{MFN tariff})$, $\ln \text{HHI}$, antidumping indicator. PPML estimated on a zero-filled HS6–dest–year grid (8.2M cells, excl. HK–Macao). Within-firm green share specification uses firm–destination and year FE (not triple-difference); see Section 5 for interpretation.

rate constructed here; it varies across destinations for the same HS6–year in 97.9% of product–year cells and is therefore a destination-facing tariff, not a Chinese import duty. Its mean does not fall after PTA entry for partner destinations, which indicates it is the MFN rather than the preferential rate. The preferential rate was not obtainable. The headline estimates do not depend on the tariff (it appears only in the “with controls” row and in the CEM matching), and whatever liberalization the agreements delivered varies at the destination–year level and is absorbed by θ_{fdt} . The residual threat runs in the conservative direction: if PTA tariff cuts fell more on green goods, the omitted preferential margin would load onto EP×green with a positive sign. The estimate is zero.

5.11 Destination-specific trends

The central falsification test asks whether allowing each destination its own linear trend in the green (and dirty) gap changes the conclusion, given that the conservative-bias argument of Section 4 implies those trends might be correlated with EP depth. Two variants bracket the answer, estimated on the collapsed panel.

The first adds destination-specific linear trends interacted with green and dirty status over the full sample period. WB results are essentially unchanged (EP×green −0.0051, bootstrap $p = 0.50$; EP×dirty −0.0082, bootstrap $p = 0.28$). The TREND green interaction flips sign

and — the only instance in the full robustness battery — survives the wild cluster bootstrap ($p = 0.012$). Taken at face value it would say environmental provisions reduce green exports. It should not be taken at face value. Unit-specific trends estimated over the full sample period absorb post-treatment dynamics and can reverse the sign of treatment coefficients when effects or cohort dynamics are present (Wolfers, 2006); the late negative drift that these trends pick up is precisely the early-cohort artifact that the Sun–Abraham estimator already identifies as noise.

The second variant estimates destination-specific slopes on *pre-treatment* years only and projects them forward before re-estimating the main specification on the detrended outcome. Every coefficient returns to an imprecise zero: $\text{TREND} \times \text{green} + 0.0074$, bootstrap $p = 0.18$; $\text{WB} \times \text{green} + 0.0168$, bootstrap $p = 0.71$; dirty interactions $p = 0.30\text{--}0.62$. That the one nominally robust coefficient in the full-period trend exercise reverses sign entirely depending on how the trends are estimated identifies it as an artifact of the trend specification. The composition null stands against destination-specific linear drifts in green demand, the most natural form of the selection-on-trends confounder.

5.12 CO₂ intensity as a continuous dirty measure

The binary dirty indicator groups HS6 products into six classic pollution-intensive sectors. A continuous intensity measure might tell a different story. I construct one from the replication data of Shapiro (2021), which reports China-specific CO₂ emission intensities for 47 EXIOBASE industries. HS6 products are mapped to ISIC3 divisions via the same HS1996–ISIC3 concordance used for the binary classification; unmatched codes (9.5% of the panel) are assigned the sample mean. The resulting measure validates against the existing trichotomy without circularity: mean intensity is 0.0012 for dirty products, 0.0008 for neutral, and 0.0006 for green, monotonic in the expected direction.

Re-estimating equation (1) on the collapsed panel with the standardized intensity in place of the binary dirty indicator gives $\text{EP} \times \text{intensity} = -0.0025$ (WB, bootstrap $p = 0.71$) and -0.0013 (TREND, asymptotic $p = 0.012$, bootstrap $p = 0.06$). The continuous measure fades under robust inference the same way the binary one does.

5.13 Alternative outcomes: quantity, unit value, and trimming

Log export value conflates shipment volume and price per unit. If EPs redirect composition without moving average prices — or vice versa — the value estimate would be an artifact of two channels canceling. Table 14 re-estimates equation (1) on the collapsed panel for three outcome variables: log export value (the main specification), log export quantity, and log

unit value.

Table 14: $EP \times \text{green}$ and $EP \times \text{dirty}$ across outcome variables (collapsed panel)

Outcome	Interaction	WB index		TREND index	
		Coeff.	Boot. p	Coeff.	Boot. p
Log export value	$\times \text{green}$	-0.0046	0.65	+0.0018	0.39
	$\times \text{dirty}$	-0.0119	0.07	+0.0004	0.86
Log export quantity	$\times \text{green}$	-0.0055	0.53	+0.0019	0.43
	$\times \text{dirty}$	-0.0115	0.25	-0.0004	0.89
Log unit value	$\times \text{green}$	$p \in [0.85, 0.95]$		$p \in [0.85, 0.95]$	
	$\times \text{dirty}$	$p \in [0.85, 0.95]$		$p = 0.16$ (TREND)	

Collapsed panel; FE $pd + dt + pt$; destination-clustered wild cluster bootstrap ($B = 9,999$). Log export value row is the baseline collapsed-panel estimate from Table 8. Unit-value coefficients are small in magnitude and reported as p -value ranges; exact coefficients in `tripledd_decomp_collapsed.csv`. Bootstrap p -values for log unit value are between 0.85 and 0.95 on all four interactions except TREND $\times$ dirty ($p = 0.16$).

No outcome registers a significant effect on either index or either product type. The null is consistent across value, quantity, and price — it is not an artifact of channels canceling in the value aggregate.

Trimming. Trimming log export value at the 1st and 99th percentiles removes 2.0% of collapsed-panel cells ($3,773,498 \rightarrow 3,698,033$); after singleton removal the estimation sample is 3,605,798. The green null is unaffected: $EP \times \text{green}$ -0.0048 (WB, bootstrap $p = 0.61$) and +0.0018 (TREND, bootstrap $p = 0.41$). The dirty margin reproduces its characteristic pattern — marginally significant under WB ($p = 0.041$), null under TREND ($p = 0.89$) — consistent with a false positive rather than a genuine effect.

6 Conclusion

China’s trade agreements did not shift the composition of its exports between 2000 and 2015. On the green margin, the finding is a *bounded null*: stable across six estimation designs and nine sample variations, robust to bootstrap and permutation inference, and bounded from above by roughly one quarter of the reference estimate in the aggregate literature. This is not a failure to reject. It is a precisely circumscribed statement about the absence of a specific effect — the kind the literature has documented elsewhere under different conditions.

On the dirty margin, the finding is a *non-result*. A coefficient that is overwhelming under asymptotic standard errors survives none of the inference battery applied to the green margin.

The leave-one-out exercise reveals why: the point estimate is stable across all 23 exclusions (-0.0097 to -0.0133), but the standard error nearly triples when either of two destinations is dropped. Precision, not the estimate, is what two specific destinations supply. An identified effect does not behave this way.

The central reading is about content. Environmental provisions affect trade when they are specific, binding, and targeted at a product category — Abman, Lundberg, and Ruta (2024) show this for deforestation, Brandi et al. (2020) for trade composition. China’s chapters of this period were thin and cooperation-heavy: the two WB sub-indices with a direct trade mechanism are perfectly collinear in-sample and appear in only three treated country-years. The policy implication is direct: a chapter in an agreement is not itself the instrument. What the chapter contains is.

Methodologically, the paper adds a cautionary case. With treatment concentrated across roughly a dozen agreements, asymptotically significant composition effects can emerge and dissolve under honest inference. Bootstrap, permutation, and leave-one-out — applied not as robustness appendices but as the primary mode of inference — should be standard practice for this class of design.

Three limitations bear noting. The preferential tariff rate was not obtainable; the headline estimates are independent of it, but the MFN rate used as a control is an imperfect proxy. A continuous-dose estimator in the spirit of Callaway, Goodman-Bacon, and Sant’Anna (2024) is not implemented; the TWFE coefficient is a weighted average of cohort-specific effects, though the permutation test is agnostic about this weighting. The extensive margin is probed at the product–destination level: new firm entry into a destination where other Chinese firms already export the same green product is not captured.

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A Equivalence of the collapsed and firm-level panels

The collapsed panel aggregates firm–HS6–destination–year observations to HS6–destination–year cells with outcome equal to the within-cell mean of log exports and weights equal to cell counts. Under fixed effects $pd + dt + pt$ the weighted cell-level regression is algebraically identical to the unweighted micro-level regression under the same fixed effects. Verification to seven significant figures: the $EP \times green$ coefficient is -0.0045685 in both; the $EP \times dirty$ coefficient is -0.011873 in both. The result is exact up to floating-point precision and holds by construction of the collapse and weights, not by approximation.

B Saturation ladder

Table 15 reports EP-depth coefficients on log export value across four fixed-effects structures, from the sparsest (firm–product–destination plus year dummies) to structures that combine two high-dimensional sets. Columns (1) and (3) are without controls, columns (2) and (4) add the non-environmental TotalDepth control. The coefficient never exceeds $+0.005$ log points per provision and reaches nominal significance only in the $fpt + pd$ structure. Two caveats apply. First, none of the four structures includes firm–destination–year effects, so the ladder does not reach the saturation of equation (1); the argument that a destination–year intercept absorbs environmental depth entirely is structural, not empirical, and is made in Section 4. Second, iterative singleton removal differs across structures, so the samples are not nested: N ranges from 22.9 million ($fpt + fpd$) to 35.6 million ($fpt + pd$). The comparison across rows therefore mixes the effect of the fixed effects with the effect of the sample.

C Sun–Abraham event study

Figure 6 and Table 16 report the interaction-weighted (L. Sun and Abraham, 2021) event study of the green and dirty versus neutral differential, estimated on the destination-year composition gap and run with `eventstudyinteract` in STATA.

A note on standard errors. The cohort shares with which `eventstudyinteract` aggregates group-specific effects are themselves estimated from the data; L. Sun and Abraham (2021) prescribe including that estimation uncertainty in the variance. `eventstudyinteract` does so; `fixest::sunab` in R treats the shares as fixed weights. The two implementations agree

Table 15: Why the level effect is not identified: saturation ladder

Fixed effects	<i>WB EP depth</i>		<i>TREND EP count</i>	
	(1) Baseline	(2) Controls	(3) Baseline	(4) Controls
<i>fpd + t</i>	0.00311 (0.00251)	0.00354 (0.00260)	0.00040 (0.00047)	0.00046 (0.00048)
<i>fpt + pd</i>	0.00463* (0.00270)	0.00476* (0.00274)	0.00101** (0.00051)	0.00105** (0.00052)
<i>fpt + fpd</i>	0.00010 (0.00326)	0.00010 (0.00329)	0.00016 (0.00061)	0.00016 (0.00060)
<i>fpd + pt</i>	0.00087 (0.00225)	0.00103 (0.00221)	0.00028 (0.00044)	0.00030 (0.00044)

Dependent variable: log export value. Each row estimates the level effect of EP depth under a different fixed-effects structure. No structure in this table includes firm–dest–year absorption; that is the specification of Table 8. The coefficient never exceeds +0.005 and reaches nominal significance only in the *fpt+pd* structure. Estimation samples differ across rows because iterative singleton removal depends on the fixed-effects structure: N = 29.5M, 35.6M, 22.9M and 29.5M from top to bottom. *f*: firm, *p*: product, *d*: destination, *t*: year. Standard errors clustered by destination. Controls in columns (2) and (4): non-environmental TotalDepth. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

on coefficients to 15 significant figures but disagree on standard errors. The degree of disagreement is informative: where a single cohort identifies a period (so there are no shares to estimate), the ratio of the two standard errors is exactly 1.00; where many cohorts contribute and their effects are heterogeneous, the ratio reaches 3 to 4. The standard errors reported here are the STATA ones, which reflect the full estimation uncertainty.

With these standard errors, no coefficient in the $[-10, +8]$ window is individually distinguishable from zero on the green or dirty margin. The ATT is -0.042 ($p = 0.27$) for the green gap and $+0.073$ ($p = 0.28$) for the dirty gap. Any apparent pre-trend in the R output (the lead at $t = -6$ had $p = 0.001$ under `fixest::sunab`) reflects the treatment of cohort-share uncertainty, not a genuine pre-trend anomaly.

D Sub-index estimates

Table 17 reports triple-difference estimates replacing the aggregate EP index with each sub-index in turn. Each row is a separate regression; the coefficients are not partial effects. See Section 5.5 for interpretation.

Table 16: Sun–Abraham interaction-weighted event study: composition gap estimates

Years since entry	Green – neutral gap		Dirty – neutral gap	
	Coeff.	p	Coeff.	p
–10	–0.0500	0.419	–0.0356	0.598
–9	–0.0128	0.881	0.0641	0.424
–8	–0.0001	0.999	–0.0478	0.436
–7	–0.0392	0.422	–0.0186	0.561
–6	0.0027	0.959	0.0466	0.335
–5	0.0022	0.961	–0.0447	0.378
–4	0.0345	0.359	–0.0210	0.538
–3	0.0021	0.938	–0.0451	0.189
–2	0.0034	0.861	–0.0080	0.832
+0	0.0116	0.504	0.0579	0.107
+1	0.0191	0.470	0.0661	0.192
+2	–0.0049	0.881	0.0765	0.241
+3	–0.0244	0.501	0.0921	0.156
+4	–0.0187	0.711	0.0906	0.181
+5	–0.0233	0.661	0.1153	0.117
+6	0.0027	0.959	0.1264	0.165
+7	–0.0320	0.541	0.1083	0.223
+8	–0.0779	0.181	0.0983	0.235
Aggregated ATT	–0.0421	0.269	0.0729	0.276

Estimated via `eventstudyinteract` in STATA (). Dependent variable: destination–year composition gap (mean log exports of green or dirty products minus mean log exports of neutral products, within destination–year cells). Standard errors propagate estimation uncertainty in cohort shares, as prescribed by L. Sun and Abraham (2021); `fixest::sunab` in R treats the shares as known weights and yields narrower standard errors — the ratio reaches 3–4 in periods identified by many discordant cohorts. Coefficients agree to 15 significant figures across implementations. Endpoint bins at $t \leq -10$ and $t \geq +8$ accumulate all observations beyond those thresholds. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 17: Provision sub-indices: which clauses have a trade mechanism?

Sub-index	×green	×dirty
<i>Mechanism-bearing sub-indices</i>		
Green goods liberalization (WB)	−0.0115 (0.0879) [0.896]	−0.0876* (0.0483) [0.071]
Green market access (TREND)	−0.0271 (0.0303) [0.372]	−0.0491** (0.0243) [0.045]
<i>Enforcement</i>		
Dispute settlement mechanism (WB)	−0.0048 (0.0394) [0.903]	0.0067 (0.0525) [0.898]
Dispute settlement mechanism (TREND)	0.0042 (0.0148) [0.778]	−0.0052 (0.0140) [0.709]
Binding obligations (TREND)	0.0022 (0.0047) [0.638]	−0.0013 (0.0037) [0.724]
<i>Placebo sub-indices (no trade mechanism)</i>		
Cooperation-only provisions (TREND)	0.0132 (0.0121) [0.275]	0.0019 (0.0108) [0.858]
Regulatory space clauses (TREND)	0.0242** (0.0099) [0.015]	0.0225*** (0.0086) [0.010]
Cells (all rows)	3,681,023	

Baseline variant, collapsed panel. Each row is a separate regression replacing the aggregate EP index with the indicated sub-index; TotalDepth controls remain included. Standard errors in parentheses; asymptotic p -values in brackets. The two mechanism-bearing WB sub-indices are perfectly collinear in-sample and non-zero in only three treated country-years (Korea from 2015, Switzerland from 2014), so their coefficients are not identified separately from each other. The regulatory space clauses, despite surviving wild cluster bootstrap, cannot be distinguished from general agreement depth (within-sample correlation with TotalDepth: 0.90); see Section 5.5. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

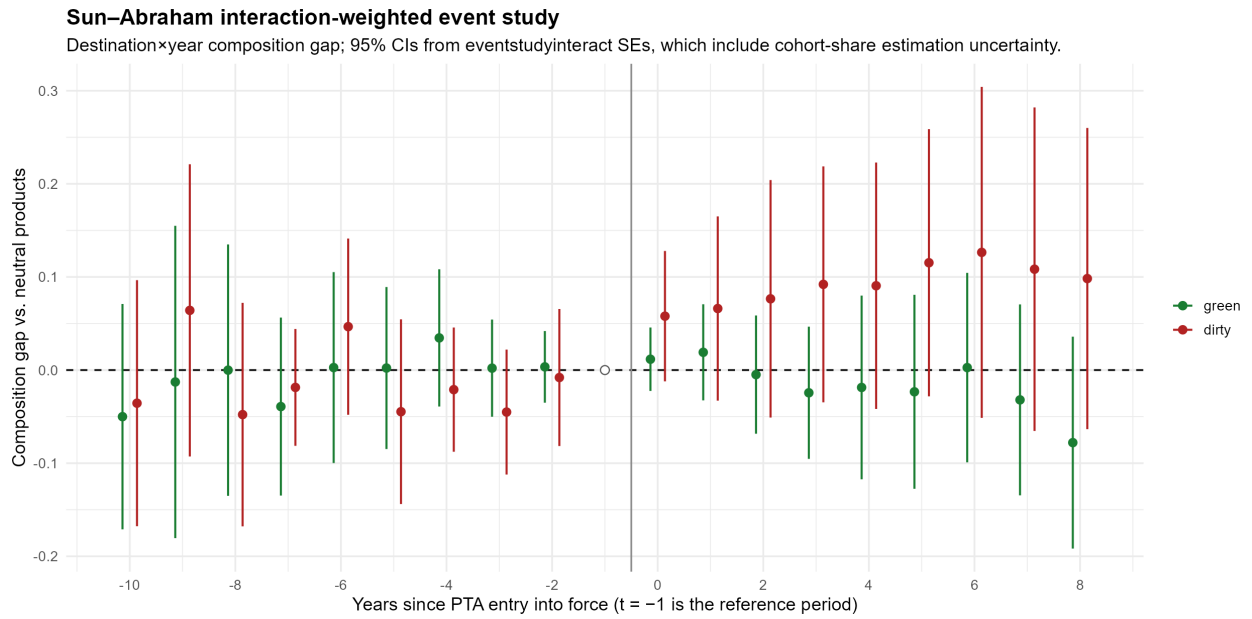


Figure 6: Sun–Abraham interaction-weighted event study: green and dirty vs. neutral gap at the destination–year level. 95% confidence bands based on `eventstudyinteract` standard errors, which account for estimation uncertainty in cohort shares. The window $[-10, +8]$ is shown; the estimates outside it rest on one or two cohorts each.